

TROIKA Model

Triple Rotor Oscillator Interlinked with K-cycle Architecture

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1. Introduction and Abstract.

We present the TROIKA model¹, a quantum system composed of three harmonic oscillators, each with an internal degree of freedom of dimension 3 (qutrit) that encodes a discrete spatial direction, $|x\rangle, |y\rangle, |z\rangle$. Its ground state is cyclically entangled,

$$|\Psi_{\text{TROIKA}}\rangle = \frac{1}{\sqrt{3}}(|xyz\rangle + |yzx\rangle + |zxy\rangle),$$

and it possesses a central property: when measuring the direction of any source, the other two are projected onto the remaining ones, always forming an orthogonal trihedron. We call this effect the **enforced trihedron**.

We build the complete formalism: we define the cyclic operators $\hat{X}_i$ ($x \rightarrow y \rightarrow z \rightarrow x$), we prove that $|\Psi_{\text{TROIKA}}\rangle$ is an eigenstate of $\hat{X}_1\hat{X}_2\hat{X}_3$ with eigenvalue +1, and we derive the directional interaction Hamiltonian

$$\hat{H}_{int}^{dir} = \frac{\hbar g}{2} (\hat{X}_1\hat{X}_2\hat{X}_3 + \text{h.c.}),$$

both by postulate and from a classical Lagrangian with cyclic symmetry. We extend the model by coupling each qutrit to its own harmonic oscillator through a local term $\hat{H}_{int}^{osc-dir} = \hbar\lambda \sum_i (\hat{X}_i\hat{a}_i^\dagger + \text{h.c.})$, which is also derived from first principles.

The complete Hamiltonian $\hat{H} = \hat{H}_0 + \hat{H}_{int}^{dir} + \hat{H}_{int}^{osc-dir}$ is expressed in matrix form using Kronecker products. By preparing the system in $|\Psi_{\text{TROIKA}}\rangle \otimes |0, 0, 0\rangle$, the fidelity exhibits two regimes: protection for $\lambda \ll g$ (stable trihedron) and coherent Rabi-like oscillations for $\lambda \sim g$, with a frequency $\Omega_{Rabi} = \sqrt{g^2 + (2\lambda)^2}$, which is compatible with an effective two-level problem.

We explore two verifications: the reduction to two sources collapses exactly into the Heisenberg XX coupling and the quantum Rabi Hamiltonian; and we sketch a continuous extension using quantum rotors in a triple-well potential, which justifies the employed discretization from first principles.

Furthermore, we possess an exact symmetry: the operator $\hat{K} = \hat{k}_1 + \hat{k}_2 + \hat{k}_3$ (mód 3) commutes with the complete Hamiltonian, decomposing the Hilbert space into three invariant sectors. The state $|\Psi_{\text{TROIKA}}\rangle$ belongs to the $K = 0$ sector, which explains its robustness against vibrational coupling and allows reducing the dynamics to a block of dimension $9(N+1)^3$ instead of $27(N+1)^3$.

Thus, the TROIKA model offers a self-contained construction of three-qutrit states with a geometric interpretation, serving as a bridge between the classical and quantum regimes, and providing a manageable numerical framework to explore entanglement robustness against vibrational degrees of freedom—with a direct connection to experimental platforms such as superconducting circuits, cold atoms, and spin systems.

¹The internal acronym used during the computational development phase was TOE (Triple Oscillator Entangled), a name that persists in the MATLAB and Python scripts for traceability reasons.

2. Isolated Qutrit: Cyclic Operator

2.1. Definition and Properties.

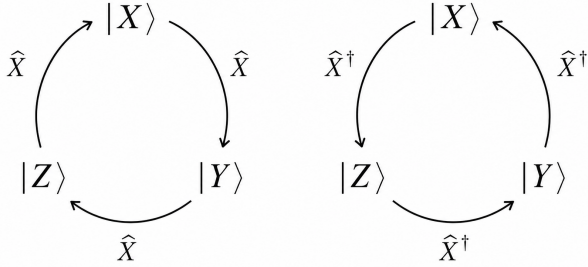
In this section, my main purpose is to define the cyclic operator $\hat{X}$ —which on some occasions I will also call the clock operator— in order to study its properties and importance in this model. Mathematically, taking into account that we will rely on the three-dimensional spatial base $\{|x\rangle, |y\rangle, |z\rangle\}$, the cyclic operator is defined as:

$$\hat{X} \equiv |y\rangle\langle x| + |z\rangle\langle y| + |x\rangle\langle z|$$

and it satisfies the following properties —hence its name:

$$\hat{X}|x\rangle = |y\rangle, \quad \hat{X}|y\rangle = |z\rangle, \quad \hat{X}|z\rangle = |x\rangle$$

Graphically, it could be symbolized by a clock-type diagram: where the operator $\hat{X}^\dagger$ is simply the adjoint (complex conjugate) of $\hat{X}$, traversing the clock in the reverse direction; mathematically:



plex conjugate) of $\hat{X}$, traversing the clock in the reverse direction; mathematically:

$$\begin{aligned} \hat{X}^\dagger &= (|y\rangle\langle x| + |z\rangle\langle y| + |x\rangle\langle z|)^\dagger = (|y\rangle\langle x|)^\dagger + (|z\rangle\langle y|)^\dagger + (|x\rangle\langle z|)^\dagger \\ &= |x\rangle\langle y| + |y\rangle\langle z| + |z\rangle\langle x| \Rightarrow \\ &\boxed{\hat{X}^\dagger = |x\rangle\langle y| + |y\rangle\langle z| + |z\rangle\langle x|} \end{aligned}$$

I will prove the properties of the operator $\hat{X}$:

$$\begin{aligned} \hat{X}|x\rangle &= (|y\rangle\langle x| + |z\rangle\langle y| + |x\rangle\langle z|)|x\rangle = |y\rangle\langle x|x\rangle + \underbrace{|z\rangle\langle y|x\rangle}_0 + \underbrace{|x\rangle\langle z|x\rangle}_0 \Rightarrow \boxed{\hat{X}|x\rangle = |y\rangle} \\ \hat{X}|y\rangle &= (|y\rangle\langle x| + |z\rangle\langle y| + |x\rangle\langle z|)|y\rangle = |y\rangle\langle x|y\rangle + \underbrace{|z\rangle\langle y|y\rangle}_1 + \underbrace{|x\rangle\langle z|y\rangle}_0 \Rightarrow \boxed{\hat{X}|y\rangle = |z\rangle} \\ \hat{X}|z\rangle &= (|y\rangle\langle x| + |z\rangle\langle y| + |x\rangle\langle z|)|z\rangle = |y\rangle\langle x|z\rangle + \underbrace{|z\rangle\langle y|z\rangle}_0 + \underbrace{|x\rangle\langle z|z\rangle}_1 \Rightarrow \boxed{\hat{X}|z\rangle = |x\rangle} \end{aligned}$$

Note: for all the calculations carried out above, it was necessary to impose from the beginning that the used directional basis —the qutrit directional basis— $\{|x\rangle, |y\rangle, |z\rangle\}$ is orthonormal, that is, it satisfies the property: $\langle i|j\rangle = \delta_i^j$, for $i = x, y, z$.

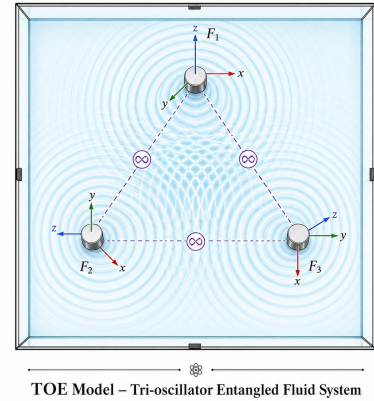
In a completely analogous way, the following relations can be proven by using the inverse cyclic operator:

$$\hat{X}^\dagger|x\rangle = |z\rangle, \quad \hat{X}^\dagger|y\rangle = |x\rangle, \quad \hat{X}^\dagger|z\rangle = |y\rangle$$

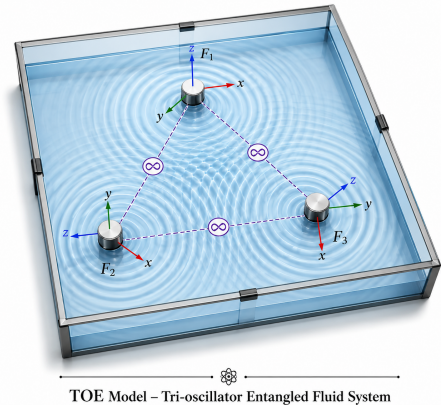
It can be seen then that the Operators $\{\hat{X}, \hat{X}^\dagger\}$ act in an analogous way to the **Weil-Heisenberg Shift Operator** in dimension 3.

2.2. Extension of the Formalism to Three Sources.

Let us move on to the interesting part: defining the three independent sources, which we will later see are directionally coupled to each other through entanglement. It is very difficult to visualize this physical system graphically in the quantum sense, but imagine a pool of water on which we have arranged three sources —for now, I am not giving importance to their geometric layout— that produce waves (this would be the classical vibrational visual interpretation). Graphically:



TOE Model – Tri-oscillator Entangled Fluid System



TOE Model – Tri-oscillator Entangled Fluid System

Where the infinity symbol between each pair of sources symbolizes that no matter the distance separating them, as soon as one turns on and the directional superposition causes its vibration direction to collapse onto one of the elements of the qutrit basis, the other sources will suffer the same fate —hence why we say they are quantum mechanically entangled.

Now, since we have three sources and three directions, each of them will have a cyclic operator and its corresponding adjoint; therefore, I define without loss of generality:

$$\begin{cases} \hat{X}_j = |y\rangle \langle x|_j + |z\rangle \langle y|_j + |x\rangle \langle z|_j \\ \hat{X}_j^\dagger = |x\rangle \langle y|_j + |y\rangle \langle z|_j + |z\rangle \langle x|_j \end{cases}, \quad \forall j = 1, 2, 3.$$

I will now assume a **pure state** of the Hilbert space $\mathcal{H}$, that is, a certain wave function $|\Psi_{TROIKA}\rangle \in \mathcal{H}$ of the form:

$$|\Psi\rangle = \frac{1}{\sqrt{3}} (|x\rangle_1 |y\rangle_2 |z\rangle_3 + |y\rangle_1 |z\rangle_2 |x\rangle_3 + |z\rangle_1 |x\rangle_2 |y\rangle_3)$$

although it can be rewritten more simply as:

$$|\Psi_{TROIKA}\rangle = \frac{1}{\sqrt{3}} (|xyz\rangle + |yzx\rangle + |zxy\rangle)$$

with both notations representing exactly the same physical-mathematical object.

First, I will prove that this wave function of the TROIKA model is an eigenstate of the total cyclic operator $\hat{X}_{123} \equiv \hat{X}_1 \hat{X}_2 \hat{X}_3$, with an eigenvalue of +1. To do this, I must take into account the directional separation by **tensor product**, that is, re-expressing:

$$|xyz\rangle \equiv |x\rangle_1 \otimes |y\rangle_2 \otimes |z\rangle_3$$

then, the state $|\Psi_{TROIKA}\rangle$ separated into the tensor products of its kets takes the form:

$$\begin{aligned} |\Psi_{TROIKA}\rangle &= \frac{1}{\sqrt{3}} (|x\rangle_1 \otimes |y\rangle_2 \otimes |z\rangle_3) + \\ &+ \frac{1}{\sqrt{3}} (|y\rangle_1 \otimes |z\rangle_2 \otimes |x\rangle_3) + \frac{1}{\sqrt{3}} (|z\rangle_1 \otimes |x\rangle_2 \otimes |y\rangle_3) \end{aligned}$$

Furthermore, we must also take into account that the joint operator $\hat{X}_1 \hat{X}_2 \hat{X}_3$ can be split into the tensor product of three unique operators:

$$\hat{X}_1 \hat{X}_2 \hat{X}_3 = \hat{X}_1 \otimes \hat{X}_2 \otimes \hat{X}_3 : \begin{cases} \hat{X}_i |k\rangle_j \sim |k+1\rangle_i \iff i = j \\ \hat{X}_i |k\rangle_j = 0, \text{ when } i \neq j. \end{cases}$$

I will now apply this operator to the state element $|xyz\rangle$ to see how it acts, and then we will generalize this behavior:

$$\begin{aligned} &\hat{X}_1 \hat{X}_2 \hat{X}_3 (|x\rangle_1 |y\rangle_2 |z\rangle_3) = \\ &= (\hat{X}_1 \otimes \hat{X}_2 \otimes \hat{X}_3) (|x\rangle_1 \otimes |y\rangle_2 \otimes |z\rangle_3) \end{aligned}$$

By the linearity properties of the tensor product, and considering that the clock operator of a particular source acts non-trivially only on directions relating to that specific qutrit, the result will be:

$$\begin{aligned} &(\hat{X}_1 |x\rangle_1) \otimes (\hat{X}_2 |y\rangle_2) \otimes (\hat{X}_3 |z\rangle_3) = \\ &= |y\rangle_1 \otimes |z\rangle_2 \otimes |x\rangle_3 \equiv |yzx\rangle \implies \boxed{\hat{X}_1 \hat{X}_2 \hat{X}_3 |xyz\rangle = |yzx\rangle} \end{aligned}$$

Following exactly the same procedure for the other two terms of the state $|\Psi_{TROIKA}\rangle$, it can be easily proven that:

$$\boxed{\hat{X}_1 \hat{X}_2 \hat{X}_3 |yzx\rangle = |zxy\rangle}, \quad \boxed{\hat{X}_1 \hat{X}_2 \hat{X}_3 |zxy\rangle = |xyz\rangle}.$$

Therefore, by letting the full cyclic operator act on the sum of these terms and multiplying by the normalization constant, I obtain:

$$\begin{aligned} \hat{X}_1 \hat{X}_2 \hat{X}_3 |\Psi_{TROIKA}\rangle &= \frac{1}{\sqrt{3}} \hat{X}_1 \hat{X}_2 \hat{X}_3 (|xyz\rangle + |yzx\rangle + |zxy\rangle) = \\ &\frac{1}{\sqrt{3}} (\underbrace{\hat{X}_1 \hat{X}_2 \hat{X}_3 |xyz\rangle}_{|yzx\rangle} + \underbrace{\hat{X}_1 \hat{X}_2 \hat{X}_3 |yzx\rangle}_{|zxy\rangle} + \underbrace{\hat{X}_1 \hat{X}_2 \hat{X}_3 |zxy\rangle}_{|xyz\rangle}) = \\ &= \frac{1}{\sqrt{3}} (|yzx\rangle + |zxy\rangle + |xyz\rangle) = |\Psi_{TROIKA}\rangle \implies \\ &\boxed{\hat{X}_1 \hat{X}_2 \hat{X}_3 |\Psi_{TROIKA}\rangle = +1 |\Psi_{TROIKA}\rangle} \end{aligned}$$

which is what we wanted to prove.

2.3. Directional Interaction Hamiltonian.

I am going to derive a quantum Hamiltonian for the qutrit direction-own source interaction. To do this, I will define a variable $\theta_i : i = 1, 2, 3$, where θ_i is a **discrete phase variable** associated with the source F_i , which can only take 3 values:

$$\theta_i \in \left\{ 0, \frac{2\pi}{3}, \frac{4\pi}{3} \right\}, \quad \forall i = 1, 2, 3$$

which symbolize the three spatial directions of the trihedron. That is,

$$\theta_i = 0 \rightarrow x, \quad \theta_i = \frac{2\pi}{3} \rightarrow y, \quad \theta_i = \frac{4\pi}{3} \rightarrow z.$$

Note: it must be taken into account that θ_i is not a real spatial angle. I use it to build a Lagrangian using trigonometric functions of θ_i , which are naturally periodic and cyclically symmetric. Furthermore, let us point out that θ_i lives only in the internal directional space $\mathcal{H}_i$.

As is standard in theoretical mechanics, I will build a classical Lagrangian and then, through the Legendre transform and by postulating a quantization correspondence, I will arrive at the quantum Hamiltonian of interest.

I want this Lagrangian to maintain symmetry under quantization and, furthermore, to make use of the variables θ_i ; the simplest one for this purpose is:

$$\mathcal{L}_{int}^{dir} = g \cos(\theta_1 + \theta_2 + \theta_3 - \phi_0)$$

where g is a directional coupling constant and ϕ_0 is a relative phase.

Now I rename $\alpha = \theta_1 + \theta_2 + \theta_3 - \phi_0$, so the Lagrangian is rewritten in a shorter form as a function of a single angle or amplitude: $\mathcal{L}_{int}^{dir} = g \cos \alpha$. Therefore, I now expand the cosine into exponentials using **Euler's identity**:

$$e^{i\alpha} = \cos \alpha + i \sin \alpha \implies e^{i\alpha} + e^{-i\alpha} = 2 \cos \alpha, \text{ hence :}$$

$$\begin{aligned} \mathcal{L}_{int}^{dir} &= g \cos \alpha = \frac{g}{2} (e^{i\alpha} + e^{-i\alpha}) = \\ &= \frac{g}{2} (e^{i(\theta_1 + \theta_2 + \theta_3 - \phi_0)} + e^{-i(\theta_1 + \theta_2 + \theta_3 - \phi_0)}) = \\ &= \frac{g}{2} (e^{i\theta_1} e^{i\theta_2} e^{i\theta_3} e^{-i\phi_0} + e^{-i\theta_1} e^{-i\theta_2} e^{-i\theta_3} e^{i\phi_0}). \end{aligned}$$

I will take the phase ϕ_0 at the reference point, that is: $\phi_0 = 0$, without loss of physical generality; thus, in the classical context:

$$\mathcal{L}_{int}^{dir} = \frac{g}{2} (e^{i\theta_1} e^{i\theta_2} e^{i\theta_3} + e^{-i\theta_1} e^{-i\theta_2} e^{-i\theta_3}),$$

which is the expression we must quantize (taking into account the discrete case). Since the Lagrangian in this case does not possess a kinetic term, it is not necessary to perform the Legendre transform: $\mathcal{L}_{int}^{dir} = H_{int}^{dir}$.

The operator $e^{i\hat{\theta}_i}$ cyclically shifts the direction state one step forward in the cycle $x \rightarrow y \rightarrow z \rightarrow x$, which is exactly what our quantum clock operator $\hat{X}_i$ does. Therefore, I postulate the **Quantization Correspondence**:

$$e^{i\theta_j} \longrightarrow \hat{X}_j, \quad e^{-i\theta_j} \longrightarrow \hat{X}_j^\dagger$$

By substituting this correspondence into the classical Lagrangian, I obtain the quantum Hamiltonian. Before expressing it, I want to make a redefinition for dimensional convenience: $g = g/\hbar$. This way, since I want the coefficients of the operators in the Hamiltonian to have energy units, I define the constant g such that it now has dimensions of frequency—this will be useful later for Rabi oscillations.

$$\hat{H}_{int}^{dir} = \frac{g\hbar}{2} (\hat{X}_1 \hat{X}_2 \hat{X}_3 + \hat{X}_1^\dagger \hat{X}_2^\dagger \hat{X}_3^\dagger)$$

The H_{int}^{dir} term models the interaction between the internal directional degrees of freedom (the qutrits) of the three sources. In the TROIKA model:

1. Each source F_i has a qutrit (direction x, y, z), and this is what this Hamiltonian couples.
2. Each source F_i also has a harmonic oscillator (vibration $n = 0, 1, 2, \dots \in \mathbb{N}$), which is an additional degree of freedom that lives in $\mathcal{H}_i^{osc}$.

Therefore, $\hat{H}_{int}^{dir}$ causes the three orientations to be correlated with each other (it is the term that leaves the state $|\Psi_{TROIKA}\rangle$ invariant).

The **free term** of the total Hamiltonian is the sum of 3 independent quantum harmonic oscillators:

$$\hat{H}_o = \sum_{i=1}^3 \hbar \omega_i \left(\hat{N}_i + \frac{1}{2} \right), \text{ with } \hat{N}_i \equiv \hat{a}_i^\dagger \hat{a}_i$$

2.4. Oscillation-Direction Interaction Hamiltonian.

The last term (the oscillatory one) is the one that connects the Direction of each source with its own oscillator. I am going to derive the expression for this oscillation Hamiltonian through a classical Lagrangian, in a similar way to how I did for $\hat{H}_{int}^{dir}$.

We already have θ_i (discrete direction phase), now I define a classical variable that describes the harmonic oscillator i . I will use the **Complex Amplitude** $\alpha_i(t)$, which is the standard variable in the action-angle description of a harmonic oscillator:

$$\alpha_i = \sqrt{\frac{m\omega_i}{2\hbar}} x_i + i \frac{p_i}{\sqrt{2m\hbar\omega_i}}$$

In terms of these variables, I will take the free Lagrangian as:

$$\mathcal{L}_o^{(i)} = \frac{i\hbar}{2} (\alpha_i^* \dot{\alpha}_i - \dot{\alpha}_i^* \alpha_i) - \hbar \omega_i |\alpha_i|^2, \text{ with } \dot{\alpha}_i = \frac{d\alpha_i}{dt}.$$

The interaction term must be:

1. **Hermitian**: real, since it is a classical Lagrangian.
2. **Local**: coupling θ_i only with its own oscillator α_i .
3. **Linear** in the oscillator amplitude,

therefore I will make use of the following expression:

$$\mathcal{L}_{int}^{osc-dir} = \hbar \lambda \sum_{i=1}^3 (\alpha_i^* e^{i\theta_i} + \alpha_i e^{-i\theta_i})$$

so I now have the **dual quantization correspondence** that I need:

$$\begin{cases} \alpha_i \rightarrow \hat{a}_i \implies \alpha_i^* \rightarrow \hat{a}_i^\dagger \\ e^{i\theta_i} \rightarrow \hat{X}_i \implies e^{-i\theta_i} \rightarrow \hat{X}_i^\dagger \end{cases}$$

where $\hat{a}$ and $\hat{a}^\dagger$ are the annihilation and creation operators, respectively:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + i \frac{\hat{p}}{\sqrt{2m\hbar\omega}}, \quad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} - i \frac{\hat{p}}{\sqrt{2m\hbar\omega}}$$

In the same way as before, since the classical Lagrangian does not depend on the time derivatives of the variables, it is not necessary to apply the Legendre transform and, finally, I obtain the **oscillator-direction interaction Hamiltonian**:

$$\hat{H}_{int}^{osc-dir} = \hbar\lambda \sum_{i=1}^3 \left(\hat{X}_i \hat{a}_i^\dagger + \hat{X}_i^\dagger \hat{a}_i \right)$$

Interpretation: when source i creates a quantum of vibration ($\hat{a}_i^\dagger$), its direction simultaneously advances one step in the cycle ($\hat{X}_i : x \rightarrow y \rightarrow z \rightarrow x$). The adjoint term ($\hat{X}_i^\dagger \hat{a}_i$) does the inverse: upon destroying a quantum of vibration, the direction goes back one step. It is a **Jaynes-Cummings-type coupling**, but between an oscillator and a qutrit instead of an oscillator and a two-level qubit.

The total Hamiltonian is then:

$$\begin{aligned} \hat{H} = & \underbrace{\hat{H}_o}_{\text{free oscillators}} + \underbrace{\hbar\lambda \sum_{i=1}^3 (\hat{X}_i \hat{a}_i^\dagger + \hat{X}_i^\dagger \hat{a}_i)}_{\text{direction-vibration coupling}} + \\ & + \underbrace{\frac{\hbar g}{2} (\hat{X}_1 \hat{X}_2 \hat{X}_3 + \hat{X}_1^\dagger \hat{X}_2^\dagger \hat{X}_3^\dagger)}_{\text{coupling between directions}} \end{aligned}$$

This means that this quantum functional, which encloses the dynamics of the TROIKA system, can be completely obtained through the Legendre transform and quantization of the following classical Lagrangian:

$$\mathcal{L} = \sum_{i=1}^3 \mathcal{L}_o^{(i)} + g \cos(\theta_1 + \theta_2 + \theta_3 - \phi_0) + \hbar\lambda \sum_{i=1}^3 (\alpha_i^* e^{i\theta_i} + \alpha_i e^{-i\theta_i})$$

3. The Quantum Harmonic Oscillator of Each Source.

3.1. Motivation and Role in the Model.

Up to this point, I have only worked with the directional degree of freedom (the qutrit) of each source F_i , and in the previous section, the transition from the classical amplitude α_i to the operators $\hat{a}_i, \hat{a}_i^\dagger$ was used without a full justification. Before moving forward towards the functional analysis of the complete system, it is convenient to pause here and properly formalize the second degree of freedom of each source: the **vibrational degree of freedom**, that is, the quantum harmonic oscillator itself.

Physically, each source F_i is not just a three-position compass: it is an oscillator that vibrates with its own natural frequency ω_i , and its vibration amplitude is quantized. The direction (qutrit) tells me where the source points; the oscillator tells me how much it is vibrating. These are two physically distinct questions, and therefore correspond to two different Hilbert spaces, which will only begin to mix once the coupling $\hat{H}_{int}^{osc-dir}$ is introduced—but first, I need to properly define the second one by itself.

3.2. Initial Classical Variables.

For the source i , I start from the canonical variables: the position x_i and the momentum p_i , with the free Lagrangian of the harmonic oscillator:

$$\mathcal{L}_0^{(i)} = \frac{1}{2} m \dot{x}_i^2 - \frac{1}{2} m \omega_i^2 x_i^2$$

This is the classical piece that, once quantized, will lead to $\hat{H}_0$. Note that here it is indeed possible to apply the Legendre transform completely and without ambiguity—unlike what happened with the discrete phase variable θ_i , which lacked a kinetic term to start from.

3.3. The Complex Amplitude: A Bridge Between the Classical and Quantum Regimes.

In the previous section, I directly introduced the variable

$$\alpha_i = \sqrt{\frac{m\omega_i}{2\hbar}} x_i + i \frac{p_i}{\sqrt{2m\hbar\omega_i}},$$

which is worth justifying now with more precision. This particular combination of x_i and p_i is the standard complex action-angle variable of a classical harmonic oscillator, and it possesses a very convenient property—under the free time evolution of the oscillator, $\alpha_i(t)$ rotates uniformly in the complex plane,

$$\alpha_i(t) = \alpha_i(0) e^{-i\omega_i t},$$

without changing its magnitude. The squared magnitude $|\alpha_i|^2$ is directly related to the classical energy of the oscillator,

$$\mathcal{H}_0^{(i)} = \hbar\omega_i |\alpha_i|^2,$$

as can be verified by substituting α_i into $\mathcal{H}_0^{(i)} = \frac{p_i^2}{2m} + \frac{1}{2} m \omega_i^2 x_i^2$ and simplifying. It is, ultimately, the natural variable to describe how much vibration is present, without the need to keep track of position and momentum separately.

3.4. Canonical Quantization.

Standard canonical quantization promotes x_i and p_i to Hermitian operators that satisfy the fundamental commutation relation

$$[\hat{x}_i, \hat{p}_i] = i\hbar,$$

and defines, from them, the creation and annihilation operators

$$\hat{a}_i = \sqrt{\frac{m\omega_i}{2\hbar}} \hat{x}_i + i \frac{\hat{p}_i}{\sqrt{2m\hbar\omega_i}}, \quad \hat{a}_i^\dagger = \sqrt{\frac{m\omega_i}{2\hbar}} \hat{x}_i - i \frac{\hat{p}_i}{\sqrt{2m\hbar\omega_i}},$$

which are nothing but the quantum version of α_i and its conjugate α_i^* —hence the correspondence $\alpha_i \rightarrow \hat{a}_i, \alpha_i^* \rightarrow \hat{a}_i^\dagger$ used previously. From the commutation relation of $\hat{x}_i$ and $\hat{p}_i$, it is directly and easily obtained that:

$$[\hat{a}_i, \hat{a}_i^\dagger] = 1, \quad [\hat{a}_i, \hat{a}_j] = [\hat{a}_i^\dagger, \hat{a}_j^\dagger] = 0 \quad (i \neq j),$$

where the second relation expresses something physically important for the TROIKA model: **the oscillators of different sources commute with each other**—there is no direct coupling between $\hat{a}_1$ and $\hat{a}_2$, for instance, anywhere in the Hamiltonian. Any correlation that appears between the oscillators of different sources must arrive indirectly, through the directional sector—this is the algebraic fact that guarantees it.

By inverting the definitions of $\hat{a}_i$ and $\hat{a}_i^\dagger$, I recover the position and momentum operators in terms of the creation and annihilation operators:

$$\hat{x}_i = \sqrt{\frac{\hbar}{2m\omega_i}} (\hat{a}_i + \hat{a}_i^\dagger), \quad \hat{p}_i = i\sqrt{\frac{\hbar m\omega_i}{2}} (\hat{a}_i^\dagger - \hat{a}_i).$$

Substituting into $\mathcal{H}_0^{(i)} = \frac{\hat{p}_i^2}{2m} + \frac{1}{2}m\omega_i^2\hat{x}_i^2$ and using the commutator $[\hat{a}_i, \hat{a}_i^\dagger] = 1$ to reorder the products $\hat{a}_i\hat{a}_i^\dagger = \hat{a}_i^\dagger\hat{a}_i + 1$, the standard form is obtained:

$$\hat{H}_0^{(i)} = \hbar\omega_i \left(\hat{a}_i^\dagger\hat{a}_i + \frac{1}{2} \right) \equiv \hbar\omega_i \left(\hat{N}_i + \frac{1}{2} \right), \quad \hat{N}_i \equiv \hat{a}_i^\dagger\hat{a}_i,$$

which is exactly the expression I already used for $\hat{H}_0 = \sum_i \hat{H}_0^{(i)}$.

3.5. The Fock Space $\mathcal{H}_i^{osc}$.

The operator $\hat{N}_i = \hat{a}_i^\dagger\hat{a}_i$ is Hermitian and therefore diagonalizable; its eigenstates $|n\rangle_i$, with $n = 0, 1, 2, \dots$, form an orthonormal basis of the vibrational space of source i , called the **Fock basis** or number basis:

$$\hat{N}_i |n\rangle_i, \quad \langle n|n'\rangle_i = \delta_{nn'}.$$

The creation and annihilation operators act on this basis in a ladder-like manner:

$$\hat{a}_i^\dagger |n\rangle_i = \sqrt{n+1} |n+1\rangle_i, \quad \hat{a}_i |n\rangle_i = \sqrt{n} |n-1\rangle_i,$$

with the particular case $\hat{a}_i |0\rangle_i = 0$ (the vacuum state cannot be destroyed any further). The entire basis is generated by applying $\hat{a}_i^\dagger$ repeatedly on the vacuum:

$$|n\rangle_i = \frac{(\hat{a}_i^\dagger)^n}{\sqrt{n!}} |0\rangle_i.$$

The vibrational space of source i is then

$$\mathcal{H}_i^{osc} = \overline{\text{span}}\{|0\rangle_i, |1\rangle_i, |2\rangle_i, \dots\},$$

a countably infinite-dimensional space (the same space that appears in any mode of the quantized electromagnetic field—the analogy with quantum optics that we already used when mentioning Jaynes-Cummings and Rabi is not accidental, it is the exact same mathematical structure).

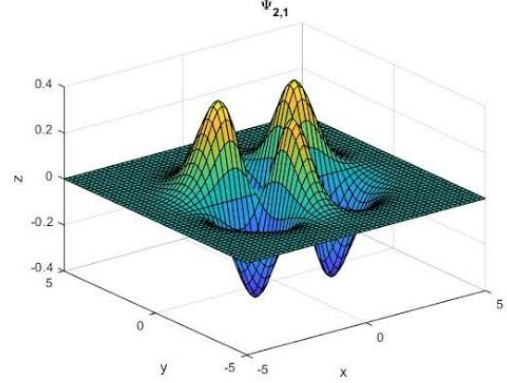
Note on the position representation: If instead of working in the abstract basis $|n\rangle_i$ one prefers a representation in terms of the physical coordinate x_i , the

same eigenstates $|n\rangle_i$ are written as the well-known Hermite functions,

$$\psi_n^{(i)}(x_i) = \langle x_i | n \rangle_i = \left(\frac{m\omega_i}{\pi\hbar} \right)^{1/4} \frac{e^{-\frac{m\omega_i}{2\hbar}x_i^2}}{\sqrt{2^n n!}} H_n \left(\sqrt{\frac{m\omega_i}{\hbar}} x_i \right)$$

where H_n is the Hermite polynomial of degree n .

An example of the representation of the probability density of such a vibrational wave function in multiple dimensions can be the following:



This representation is not necessary for the matrix and numerical development that follows—I will work exclusively in the abstract basis $|n\rangle_i$ —but it is worth mentioning because it provides concrete physical meaning to the vibration: $|\psi_n^{(i)}(x_i)|^2$ is literally the probability distribution of finding source i displaced a distance x_i from its equilibrium position when it has n quanta of vibrational excitation.

3.6. The Extended Hilbert Space of Each Source.

With the directional sector $\mathcal{H}_i = \text{span}\{|x\rangle_i, |y\rangle_i, |z\rangle_i\}$ (dimension 3) and the vibrational sector $\mathcal{H}_i^{osc}$ that I have just defined (infinite dimension), the complete Hilbert space associated with source F_i is the tensor product of both:

$$\mathcal{H}_i^{total} = \mathcal{H}_i \otimes \mathcal{H}_i^{osc}.$$

A natural basis for this space is the product basis

$$|n, a\rangle_i \equiv |n\rangle_i \otimes |a\rangle_i, \quad n \in \{0, 1, 2, \dots\}, \quad a \in \{x, y, z\},$$

which is the one I will use from now on to describe the state of each source separately, and it is also the one underlying the expansion

$$\sum c_{n_1 n_2 n_3}^{abc}(t) |n_1, a\rangle_1 |n_2, b\rangle_2 |n_3, c\rangle_3$$

which will be introduced in order to formulate the system of coupled equations.

It is important to emphasize, before moving forward, that $|n, a\rangle_i$ is not a mixed state between direction and vibration in any special physical sense: it is simply the joint specification of two independent quantum numbers (n , quanta of vibration; a , direction), just as in the hydrogen atom one jointly specifies (n, l, m) .

The independence of both sectors is reflected in the fact that any directional operator (such as $\hat{X}_i$) acts trivially on the factor $|n\rangle_i$ (leaving it unchanged), and any vibrational operator ($\hat{a}_i, \hat{a}_i^\dagger, \hat{N}_i$) acts trivially on the factor $|a\rangle_i$:

$$\begin{cases} \hat{X}_i(|n\rangle_i \otimes |a\rangle_i) = |n\rangle_i \otimes (\hat{X}_i |a\rangle_i) \\ \hat{a}_i^\dagger(|n\rangle_i \otimes |a\rangle_i) = (\hat{a}_i^\dagger |n\rangle_i) \otimes |a\rangle_i \end{cases}$$

It is precisely $\hat{H}_{int}^{osc-dir} = \hbar\lambda \sum_i (\hat{X}_i \hat{a}_i^\dagger + \hat{X}_i^\dagger \hat{a}_i)$ that constitutes the only operator in the model that acts in a genuinely joint manner on both factors at the same time—neither purely on $\mathcal{H}_i$ nor purely on $\mathcal{H}_i^{osc}$, but mixing both—and this is exactly why it is the term responsible for all the direction-vibration entanglement that will be observed dynamically.

3.7. The Total Hilbert Space of the Extended TROIKA System.

By assembling the three sources, the total Hilbert space of the extended TROIKA model becomes

$$\mathcal{H}_{TROIKA} = \bigotimes_{i=1}^3 \mathcal{H}_i^{total} = \bigotimes_{i=1}^3 (\mathcal{H}_i \otimes \mathcal{H}_i^{osc})$$

with the complete product basis

$$|n_1, a\rangle_1 \otimes |n_2, b\rangle_2 \otimes |n_3, c\rangle_3, \quad \begin{cases} a, b, c \in \{x, y, z\}, \\ n_1, n_2, n_3 \in \{0, 1, 2, \dots\}. \end{cases}$$

In practice, for any explicit calculation (whether numerical or analytical in finite dimension), I will truncate the vibrational space of each source to a maximum number of quanta n_{max} , so that $\mathcal{H}_i^{osc} \rightarrow \text{span}\{|0\rangle_i, \dots, |n_{max}\rangle_i\}$, with $\dim \mathcal{H}_i^{osc} = n_{max} + 1$. This is the basis on which the entire functional analysis of the complete system will rest in the next section: how many degrees of freedom there are, how the total dimension is organized, and how the different sectors interact (or not) with each other.

4. Functional Analysis of the Extended TROIKA System

4.1. Motivation and Approach.

We now have all the formal ingredients of the model at our disposal: the directional space $\mathcal{H}_i \cong \mathbb{C}^3$ of each source, its vibrational space $\mathcal{H}_i^{osc}$, the extended space $\mathcal{H}_i^{total} = \mathcal{H}_i \otimes \mathcal{H}_i^{osc}$, and the complete Hamiltonian

$$\hat{H} = \hat{H}_0 + \hat{H}_{int}^{dir} + \hat{H}_{int}^{osc-dir}.$$

Before proceeding to its numerical study, I dedicate this section to characterizing the internal architecture of $\mathcal{H}_{TROIKA}$: its dimension, its natural decomposition into invariant subspaces, and—above all—the exact conserved quantities that this decomposition reveals.

The goal is for the phenomenology that will be observed numerically in the next section (regimes of protection, Rabi-like oscillations) to be anticipated here by purely algebraic arguments, rather than appearing as a purely empirical finding of the code.

4.2. Layered Structure of the Hilbert Space.

The total space of the model,

$$\mathcal{H}_{TROIKA} = \bigotimes_{i=1}^3 (\mathcal{H}_i \otimes \mathcal{H}_i^{osc}),$$

allows, by the associativity and commutativity of the tensor product, an equivalent regrouping that separates the nature of the degrees of freedom instead of the source to which they belong:

$$\mathcal{H}_{TROIKA} \cong \mathcal{H}_{dir} \otimes \mathcal{H}_{vib}, \quad \begin{cases} \mathcal{H}_{dir} \equiv \bigotimes_{i=1}^3 \mathcal{H}_i \quad (\dim = 27), \\ \mathcal{H}_{vib} \equiv \bigotimes_{i=1}^3 \mathcal{H}_i^{osc}. \end{cases}$$

This second factorization is not a mere notational artifact: it is the one that exhibits the physical structure of the problem. The state $|\Psi_{TROIKA}\rangle$ is, by construction, a vector of $\mathcal{H}_{dir} \otimes |0, 0, 0\rangle \subset \mathcal{H}_{TROIKA}$, and the operators $\hat{H}_0$ and $\hat{H}_{int}^{dir}$ act as $\hat{1}_{dir} \otimes \hat{H}_0^{vib}$ and $\hat{H}_{int}^{dir} \otimes \hat{1}_{vib}$, respectively—each one trivially respecting the factorization—while

$$\hat{H}_{int}^{osc-dir} = \hbar\lambda \sum_{i=1}^3 (\hat{X}_i \otimes \hat{a}_i^\dagger + \hat{X}_i^\dagger \otimes \hat{a}_i)$$

is, in a precise sense, the only generator of entanglement between $\mathcal{H}_{dir}$ and $\mathcal{H}_{vib}$ contained in the model.

4.3. System Dimension and Truncation.

By truncating each Fock space to $N + 1$ levels ($n_i = 0, \dots, N$), the dimension of each local factor is $\dim \mathcal{H}_i^{total} = 3(N + 1)$, and by the multiplicativity of the dimension under the tensor product,

$$\dim \mathcal{H}_{TROIKA} = [3(N + 1)]^3 = 27(N + 1)^3$$

The case $N = 1$ used in the numerical simulation corresponds to $\dim \mathcal{H}_{TROIKA} = 216$; the cubic growth in N is the formal argument that justifies resorting to numerical methods as soon as a non-negligible vibrational resolution is required, in contrast with the exact analytical solvability of the $N = 0$ cases (purely directional, 27 dimensions) that were treated in the preceding sections.

4.4. Cyclic Symmetry with Two Distinct $\mathbb{Z}^3$ Groups.

This model possesses two cyclic symmetries of different origin, whose distinction is essential and should be formalized through the corresponding group action.

(i) **Qudit symmetry** (internal $\mathbb{Z}_3$): The cyclic group generated by $\hat{X}_i$ acts on each factor $\mathcal{H}_i$ individually, as an automorphism of the basis $\{|x\rangle, |y\rangle, |z\rangle\}_i$. This is the symmetry previously used to prove that $\hat{X}_1\hat{X}_2\hat{X}_3|\Psi_{TROIKA}\rangle = |\Psi_{TROIKA}\rangle$.

(ii) **Source permutation symmetry** (external $\mathbb{Z}_3$): Let S_3 be the symmetric group over the set of indices $\{1, 2, 3\}$, acting on $\mathcal{H}_{TROIKA}$ through the representation operator

$$\hat{P}_\sigma(|\psi_1\rangle \otimes |\psi_2\rangle \otimes |\psi_3\rangle) = |\psi_{\sigma^{-1}(1)}\rangle \otimes |\psi_{\sigma^{-1}(2)}\rangle \otimes |\psi_{\sigma^{-1}(3)}\rangle$$

where $\sigma \in S_3$.

Restricted to the cyclic subgroup $\mathbb{Z}_3 \subset S_3$ generated by $\rho = (1, 2, 3)$, it is straightforward to verify —by a simple rearrangement of the three summands that compose $|\Psi_{TROIKA}\rangle$ — that

$$\hat{P}_\rho |\Psi_{TROIKA}\rangle = |\Psi_{TROIKA}\rangle,$$

an invariance that is structurally independent of the one established in (i). Under the physical condition $\omega_1 = \omega_2 = \omega_3 \equiv \omega$ (which we will adopt in the first simulation), it also holds that $[\hat{H}, \hat{P}_\rho] = 0$: the external cyclic symmetry is exact for the full Hamiltonian, not just for its directional sector. This is the formal reason for the exact overlap observed between $\langle \hat{N}_1(t) \rangle$, $\langle \hat{N}_2(t) \rangle$, and $\langle \hat{N}_3(t) \rangle$: such a degeneracy is a direct consequence of $\hat{H} \in \text{End}(\mathcal{H}_{TROIKA})^{\mathbb{Z}_3}$, the commutant of the external cyclic representation.

4.5. Spectral Decomposition of the Directional Sector.

The operator $\hat{X}_i$, which is unitary in $\mathcal{H}_i \cong \mathbb{C}^3$, is diagonalized in the basis of its eigenvectors $\{|\phi_k\rangle_i\}_{k=0}^2$, where

$$\hat{X}_i |\phi_k\rangle_i, \quad \omega = e^{2\pi i/3}$$

are the irreducible characters of $\mathbb{Z}_3$. In the corresponding product basis, $\hat{X}_1\hat{X}_2\hat{X}_3$ acts diagonally with eigenvalue $\omega^{k_1+k_2+k_3}$, which induces the direct sum decomposition on $\mathcal{H}_{dir}$:

$$\mathcal{H}_{dir} = \bigoplus_{K=0}^2 \mathcal{H}_{dir}^{(K)}, \quad \text{where we have:}$$

$$\mathcal{H}_{dir}^{(K)} \equiv \text{span}\{|\phi_{k_1}\phi_{k_2}\phi_{k_3}\rangle : k_1+k_2+k_3 \equiv K \pmod{3}\}$$

with $\dim \mathcal{H}_{dir}^{(K)} = 9$ for each K (consistent with $27 = 9 + 9 + 9$). Each $\mathcal{H}_{dir}^{(K)}$ is an eigenspace of $\hat{H}_{int}^{dir}$ with energy

$$E^{(K)} = \hbar g \cos\left(\frac{2\pi K}{3}\right), \quad E^{(0)} = \hbar g, \quad E^{(1)} = E^{(2)} = -\frac{\hbar g}{2},$$

thus obtaining the spectrum, where $|\Psi_{TROIKA}\rangle \in \mathcal{H}_{dir}^{(0)}$ occupies the extremal value.

4.6. K as an Exact Invariant of the Full Generator of Dynamics.

The central result of this section is that the label K is not just a good quantum number for $\hat{H}_{int}^{dir}$, but a conserved quantity of the complete Hamiltonian. In the $\{|\phi_k\rangle_i\}$ basis, the operator $\hat{X}_i$ acts by multiplication (not by shift) on the index k_i ; consequently, the term $\hat{X}_i\hat{a}_i^\dagger + \hat{X}_i^\dagger\hat{a}_i$ modifies $n_i \rightarrow n_i \pm 1$ leaving k_i —and therefore K — invariant. Formally,

$$[\hat{H}, \hat{K}] = 0, \quad \hat{K} \equiv (\hat{k}_1 + \hat{k}_2 + \hat{k}_3) \pmod{3}$$

for any values of the couplings g, λ . This identity induces the decomposition on the complete extended Hilbert space:

$$\mathcal{H}_{TROIKA} = \bigoplus_{K=0}^2 \mathcal{H}_{dir}^{(K)} \otimes \mathcal{H}_{vib}, \quad \text{where in general:}$$

$$\dim(\mathcal{H}_{dir}^{(K)} \otimes \mathcal{H}_{vib}) = 9(N+1)^3,$$

into three mutually invariant subspaces under the unitary evolution generated by $\hat{H}$: no matrix element of $\hat{H}$ connects different K sectors. Since the initial state is prepared in $|\Psi_{TROIKA}\rangle \otimes |0, 0, 0\rangle \in \mathcal{H}_{dir}^{(0)} \otimes \mathcal{H}_{vib}$, the entire dynamics of the system is exactly confined to the $K = 0$ block —with dimension $9(N+1)^3$, which is remarkably smaller than the nominal dimension $27(N+1)^3$ of the total space—. This confinement is neither an approximation nor an empirical consequence of the chosen initial conditions: it is a necessary consequence of $[\hat{H}, \hat{K}] = 0$, and it provides the structural explanation for the robustness of the $K = 0$ sector against the vibrational coupling that will be numerically quantified later.

4.7. Effective Reduction Within the Invariant Sector.

Within a fixed K block, the condition $k_1+k_2+k_3 \equiv K \pmod{3}$ determines k_3 as a function of (k_1, k_2) , so that the $\mathcal{H}_{dir}^{(K)} \otimes \mathcal{H}_{vib}$ block is unitarily equivalent to a system of three linearly coupled oscillators with relative phases $\omega^{k_1}, \omega^{k_2}, \omega^{k_3(k_1, k_2)}$ fixed by K —the precise structure that, in the two-source limit, collapses exactly to the quantum Rabi Hamiltonian, which I will prove later—. This argument anticipates that the relevant energy scale for the fidelity dynamics of $|\Psi_{TROIKA}\rangle$ must combine the spectral gap $E^{(0)} - E^{(1,2)} = \frac{3}{2}\hbar g$ with the coupling scale $\hbar\lambda$, in the characteristic form of a generalized Rabi formula —consistent with the $\Omega_{Rabi} = \sqrt{g^2 + (2\lambda)^2}$ law that will be obtained by numerical fitting—.

4.8. Algebraic Observation: The Discrete Weil-Heisenberg Pair.

It is worth placing $\hat{X}_i$ within a broader and already standardized algebraic framework in the theory of finite-dimensional quantum systems (qudits): it is the shift operator of the discrete Weil-Heisenberg pair in $d = 3$, commonly denoted as X in this literature. Along with the associated phase operator,

$$\hat{Z}_i \equiv |x\rangle\langle x|_i + \omega |y\rangle\langle y|_i + \omega^2 |z\rangle\langle z|_i$$

it satisfies the canonical Weyl commutation relation,

$$\hat{X}_i \hat{Z}_i = \omega \hat{Z}_i \hat{X}_i,$$

which generates the complete discrete Heisenberg-Weyl group over $\mathbb{C}^3$. This identification frames the clock operator —previously introduced in an ad hoc manner— within a standard algebraic structure, and suggests a natural path to extend the TROIKA model to the general formalism of qudit stabilizers.

Term	Support	Invariant
$\hat{H}_0$	$\mathcal{H}_{\text{vib}}$	n_1, n_2, n_3
$\hat{H}_{\text{int}}^{\text{dir}}$	$\mathcal{H}_{\text{dir}}$	k_1, k_2, k_3
$\hat{H}_{\text{int}}^{\text{osc-dir}}$	$\mathcal{H}_i \otimes \mathcal{H}_i^{\text{osc}}$	$K = (k_1 + k_2 + k_3) \bmod 3$

5. Numerical Simulations of Interest.

5.1. Simulation of the Extended TROIKA Model Dynamics.

5.1.1. Executive Summary.

The first script simulates the time dynamics of the extended TROIKA model in a Hilbert space of dimension $N_{\text{total}} = 1728$. It calculates the evolution of the entangled initial state $|\Psi_0\rangle = |\Psi_{\text{TROIKA}}\rangle \otimes |0, 0, 0\rangle$ under the total Hamiltonian to evaluate the **Fidelity** $F(t) = |\langle\Psi_0|\Psi(t)\rangle|^2$ in three coupling regimes ($\lambda/g \in \{0.05, 0.30, 0.80\}$), directly comparing it with the evolution of the expected value of the total number of vibrational quanta $\langle\hat{N}_{\text{tot}}\rangle(t)$.

5.1.2. Analytical Foundation.

The algorithm constructs the global Hilbert space through iterated tensor products of individual source subspaces $\mathcal{H}_i^{\text{total}} = \mathcal{H}_i^{\text{dir}} \otimes \mathcal{H}_i^{\text{osc}}$ ($\dim = 3 \times 4 = 12$ per source):

$$\mathcal{H}_{\text{TROIKA}} = \bigotimes_{i=1}^3 (\mathcal{H}_i^{\text{dir}} \otimes \mathcal{H}_i^{\text{osc}}), \quad \dim(\mathcal{H}_{\text{TROIKA}}) = 12^3 = 1728.$$

The local operators are extended to the total space, preserving their inter-source commutativity, by using Kronecker products (**kron**). The dynamics are governed by the complete Hamiltonian:

$$\begin{aligned} \hat{H} = & \sum_{i=1}^3 \hbar \omega_i \left(\hat{N}_i + \frac{1}{2} \hat{\mathbb{I}} \right) + \frac{\hbar g}{2} \left(\hat{X}_1 \hat{X}_2 \hat{X}_3 + \hat{X}_1^\dagger \hat{X}_2^\dagger \hat{X}_3^\dagger \right) + \\ & + \hbar \lambda \sum_{i=1}^3 \left(\hat{X}_i \hat{a}_i^\dagger + \hat{X}_i^\dagger \hat{a}_i \right). \end{aligned}$$

To solve the time-independent Schrödinger equation $\hat{H} |\psi_n\rangle = E_n |\psi_n\rangle$, the script exactly diagonalizes the Hamiltonian in the truncated Fock basis ($n_{\text{max}} = 3$):

$$\hat{H} = \hat{V} \hat{D} \hat{V}^\dagger.$$

The evolution of the state vector at time t is obtained by projecting the initial state onto the eigenvector basis ($c_n = \langle\psi_n|\Psi_0\rangle$) and applying the dynamic phase factors:

$$|\Psi(t)\rangle = \sum_{n=1}^{\dim} c_n e^{-iE_n t/\hbar} |\psi_n\rangle$$

From $|\Psi(t)\rangle$, two primary Hermitian observables are evaluated:

1. **Fidelity** with respect to the prepared state:

$$F(t) = |\langle\Psi_0|\Psi(t)\rangle|^2$$

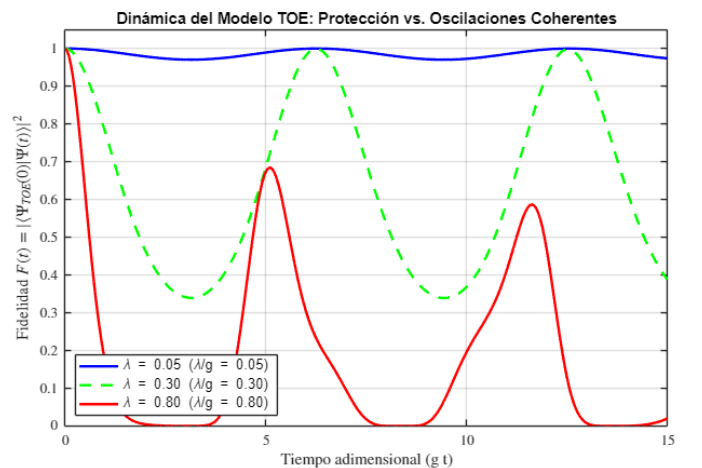
2. **Total population of vibrational excitations:**

$$\langle\hat{N}_{\text{tot}}\rangle(t) = \langle\Psi(t)| \left(\hat{N}_1 + \hat{N}_2 + \hat{N}_3 \right) |\Psi(t)\rangle.$$

5.1.3. Representation and Physical Relevance.

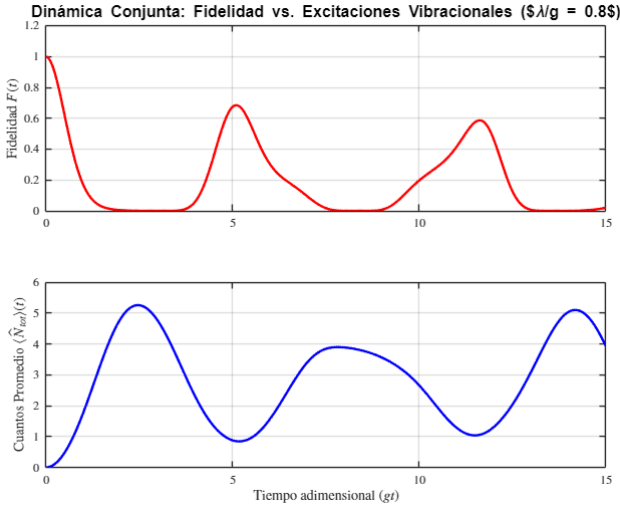
Two fundamental figures are plotted:

1. **Fidelity** $F(t)$ for multiple values of λ/g :



It shows the transition between the frozen/protected subspace regime ($\lambda \ll g$, where $F \approx 1$ in a quasi-stationary manner) and the coupled Rabi oscillations regime ($\lambda \sim g$, where periodic collapses and revivals appear).

2. Joint dynamics $F(t)$ vs. $\langle \hat{N}_{tot} \rangle(t)$:



It shows an exact anti-correlation relationship between the loss of fidelity in the directional sector and the increase in the average number of bosons created in the sources.

This representation is crucial because it demonstrates that the loss of superposition of the enforced trihedron $|\Psi_{TROIKA}\rangle$ is not due to a decoherence process caused by environmental noise, but rather to a unitary and reversible transfer of quantum information between the directions of the qutrits and the vibrational degrees of freedom of the sources.

5.2. Simulation of Hybrid Entanglement Dynamics and Von Neumann Entropy.

5.2.1. Executive Summary.

The second script simulates the hybrid entanglement dynamics between the directional degrees of freedom and the bosonic vibration modes under a strong coupling regime ($\lambda/g = 0.80$). To do this, we calculate at each time instant the reduced density operator of the directional subspace $\hat{\rho}_{dir}(t)$ through the partial trace over the oscillators, and evaluate the von Neumann entropy $S(\hat{\rho}_{dir})$, directly comparing its time evolution with the loss of fidelity $F(t)$.

5.2.2. Mathematical and Analytical Foundation.

The pure global state $|\Psi(t)\rangle \in \mathcal{H}_{dir} \otimes \mathcal{H}_{vib}$ possesses a pure density matrix $\hat{\rho}(t) = |\Psi(t)\rangle \langle \Psi(t)|$. Since the excitations of the vibrational field cannot be measured in isolation, the physics is projected onto the directional subspace of the qutrits ($\dim = 27$) by averaging out the oscillatory degrees of freedom ($\dim = 64$) via the partial trace:

$$\hat{\rho}_{dir}(t) = \text{Tr}_{vib} (|\Psi(t)\rangle \langle \Psi(t)|) =$$

$$= \sum_{n_1, n_2, n_3} \langle n_1, n_2, n_3 | \Psi(t) \rangle \langle \Psi(t) | n_1, n_2, n_3 \rangle.$$

Mathematically, the code implements this operation in an extremely efficient way by reshaping the state vector $|\Psi(t)\rangle \in \mathbb{C}^{1728}$ into a matrix $\Psi_{mat} \in \mathbb{C}^{27 \times 64}$, where the matrix contraction product performs the partial trace exactly:

$$\hat{\rho}_{dir} = \Psi_{mat} \Psi_{mat}^\dagger.$$

The von Neumann entropy of the directional subspace quantifies the degree of quantum mixing (and, therefore, the entanglement entropy between the subspaces):

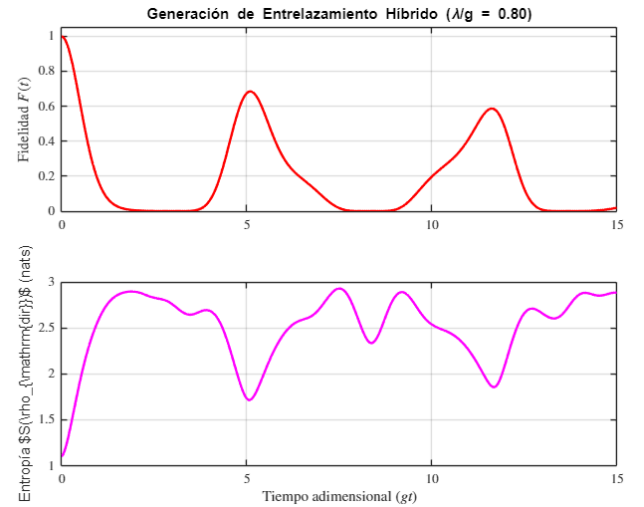
$$S(\hat{\rho}_{dir}) = -\text{Tr} (\hat{\rho}_{dir} \ln \hat{\rho}_{dir}) = -\sum_i p_i \ln(p_i)$$

where $\{p_i\}$ are the non-zero eigenvalues of the reduced density operator $\hat{\rho}_{dir}$ ($p_i > 0, \sum p_i = 1$).

5.2.3. Representation and Physical Relevance.

Two overlaid panels are plotted as a function of the dimensionless time gt :

1. **Upper panel ($F(t)$):** It shows the periodic loss and recovery of fidelity of the prepared initial state.
2. **Lower panel ($S(\hat{\rho}_{dir})$):** It shows the evolution of the degree of entropy of the directional sector.



When the state starts at $t = 0$, the directional subspace is in a pure state ($S = 0$). As the coupling λ acts, $S(\hat{\rho}_{dir})$ grows towards positive values, quantitatively demonstrating that the directional subspace undergoes a statistical mixing induced by the entanglement with the vibrational matrix.

The entropy $S(\hat{\rho}_{dir})$ reaches its local maxima exactly at the minima of the fidelity $F(t)$, and approaches zero again when the fidelity recovers during the **Rabi revivals**. This confirms that the system is not experiencing an irreversible (Markovian) decoherence process, but rather a coherent and fully reversible dynamic entanglement between qutrits and oscillators.

Rabi Revivals: In the dynamics of coupled quantum systems —such as the Jaynes-Cummings model or this qutrit-oscillator system— the probability of finding the system in its initial state (the fidelity $F(t)$) oscillates harmonically. When quantum information ‘flows’ from the directional sector toward the oscillators, the fidelity drops and the entanglement entropy increases. However, being a closed system of finite dimension with unitary evolution, information is not destroyed: it periodically returns to the qutrits. A Rabi revival is precisely that instant in which the phases of the quantum states realign coherently, the entropy $S(\hat{\rho}_{dir})$ collapses back to zero, and the fidelity $F(t)$ is almost fully recovered, demonstrating the oscillatory and cyclic nature of the exchange of energy and information.

Irreversible Markovian Decoherence: This refers to the process by which a quantum system definitively loses its coherence (its quantum phases and interference capability) when interacting with a macroscopic environment or “thermal bath” with infinite degrees of freedom. Under the Markovian approximation (zero memory of the environment), information dissipates into the surroundings as if through a one-way channel with no return (heat leakage/thermal noise). The reduced density matrix becomes an irreversible statistical mixture where entropy grows monotonically until it stabilizes at a maximum value. The fact that my model is non-Markovian emphasizes that the drop in fidelity is not due to environmental noise or thermal dissipation, but to a pure, unitary, and fully reversible quantum entanglement with the internal vibrational matrix of the model itself.

5.3. Quasiprobability Simulation Through Wigner Functions.

5.3.1. Executive Summary.

This script reconstructs the quasiprobability distribution through the Wigner function $W(x_1, p_1)$ in the phase space (x_1, p_1) for the first oscillator at the time instant $t = 2.5$ (close to the fidelity minimum, as seen in the previous plots). It extracts a reduced density matrix $\hat{\rho}_{osc1}(t)$ by tracing out the remaining 5 subspaces and evaluates the harmonic oscillator wave functions to map deviations from classical Gaussian states and regions of quantum negativity.

5.3.2. Mathematical and Analytical Foundation.

To isolate the state of a single vibrational degree of freedom, the global vector of dimension $1728(n_{max} = 5)$ is converted into a 6th-order tensor via space bijection:

$$\mathcal{H}_{TROIKA} \cong \mathcal{H}_1^{osc} \otimes (\mathcal{H}_1^{dir} \otimes \mathcal{H}_2^{dir} \otimes \mathcal{H}_2^{osc} \otimes \mathcal{H}_3^{dir} \otimes \mathcal{H}_3^{osc}).$$

Through permutation and matrix contraction indices, the code performs the partial trace over the other 5 subspaces (the 3 directions and oscillators 2 and 3),

reducing the state to a matrix $\hat{\rho}_{osc1} \in \mathbb{C}^{6 \times 6}$:

$$\hat{\rho}_{osc1} = Tr_{\dim_{1,2,3} \, osc_{2,3}} (|\Psi(t)\rangle \langle \Psi(t)|) = \sum_{n,m=0}^{n_{max}} \rho_{n,m} |n\rangle \langle m|.$$

The Wigner function $W(x, p)$ maps the density operator from the discrete Fock representation to the continuous phase space (x, p) through the Wigner-Weyl transform:

$$W(x, p) = \frac{1}{2\pi\hbar} \int_{-\infty}^{\infty} \langle x - \frac{y}{2} | \hat{\rho}_{osc1} | x + \frac{y}{2} \rangle e^{ipy/\hbar} dy.$$

Substituting the decomposition in the Fock basis, the integral expands in terms of the position wave functions $\psi_n(x) = \langle x | n \rangle$, which involve the Hermite polynomials $H_n(x)$:

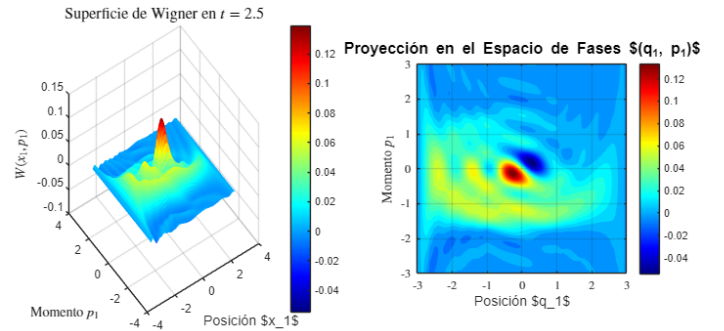
$$\psi_n(x) = \frac{1}{\pi^{1/4} \sqrt{2^n n!}} H_n(x) e^{-x^2/2},$$

$$W(x, p) = \frac{1}{2\pi} \sum_{n,m=0}^{n_{max}} \rho_{n,m} \int_{-\infty}^{\infty} \psi_m\left(x + \frac{y}{2}\right) \psi_n^*\left(x - \frac{y}{2}\right) e^{-2ipy} dy$$

5.3.3. Representation and Physical Relevance.

Two complementary panels are generated for the time instant $t = 2.5$:

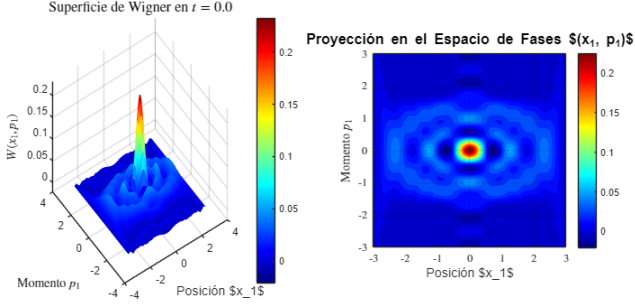
1. **Left panel (3D Surface):** It shows the three-dimensional topography of $W(x_1, p_1)$ over the phase plane.
2. **Right panel (2D Contour Map):** It shows the 2D projection on the (x_1, p_1) plane.



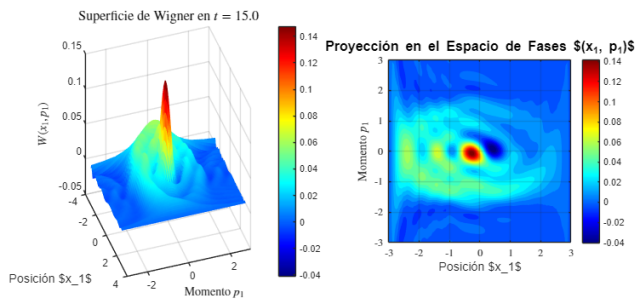
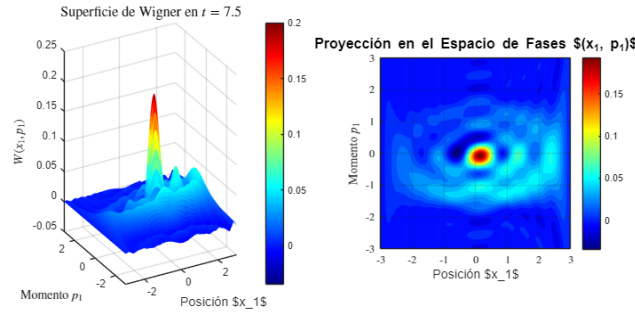
For the initial vacuum state $|0, 0, 0\rangle$, the Wigner function is a symmetric, positive Gaussian centered at the origin. At $t = 2.5$, the non-linear coupling with the qutrit subspace deforms the distribution, potentially generating regions with negative values ($W(x, p) < 0$). The Wigner negativity is the standard metric that formally proves that the generated vibrational state is neither a classical statistical mixture nor a coherent state, but a non-classical quantum state.

The appearance of wave-like patterns in $W(x_1, p_1)$ reflects the generation of macroscopic superpositions (Schrödinger cat-like states or squeezed states) induced by the coherence transfer from the directional subspace.

The Wigner function plots at the **maxima of the Fidelity function** possess very similar shapes, which is why I decided to represent only one of them (referring to the time instants $t \in \{0, 5, 12.15\}$):



For values where the fidelity is minimal ($t \in \{2.5, 7.5, 15\}$), the representations are all very similar to the one already presented for the time instant $t = 2.5$:



It is then observed that at the points of maximum fidelity, the Wigner function recovers its well-behaved Gaussian symmetry, whereas at the fidelity minima, the negativity wells ($W(x, p) < 0$) and interference patterns clearly appear.

This demonstrates in a visual and undeniable way that the loss of fidelity is not decoherence due to noise (which would erase the interferences, destroying the negativity), but rather the active generation of an entangled, non-classical state hybridized between the qutrit and the oscillator.

6. Limiting Case: Simplification of the TROIKA Model to Two Sources.

6.1. Motivation.

The TROIKA model has been constructed, up to this point, on three sources and one qutrit per source—the minimal choice that allows for a genuine orthogonal trihedron. It is natural, and methodologically valuable, to ask what happens if the number of sources and directions is reduced to two. This reduction is not a mere exercise in simplification: as will be shown, the model collapses exactly into two structures already extensively studied in quantum physics—the Heisenberg exchange coupling and the quantum Rabi Hamiltonian—which constitutes a non-trivial cross-verification of the general construction. The fact that a new model reproduces known and well-established results in its simplest limit is evidence that its structure is not arbitrary.

6.2. The 2-Dimensional Cyclic Operator.

With only two directions $\{|x\rangle, |y\rangle\}$, the clock operator reduces to

$$\hat{X} \equiv |y\rangle\langle x| + |x\rangle\langle y|,$$

which in its own directional basis $\{|x\rangle, |y\rangle\}$ is exactly the Pauli matrix σ_x :

$$\hat{X} \longrightarrow \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Unlike the $d = 3$ case, here $\hat{X}^\dagger = \hat{X}$ and $\hat{X}^2 = \hat{1}$ hold true: the two-element cyclic operator is simultaneously Hermitian and unitary, a property that has no analogue in $d \geq 3$ and remarkably simplifies all subsequent developments.

6.3. The Two-Source TROIKA State: A Bell State.

The analogue of $|\Psi_{TROIKA}\rangle$ for two sources is the cyclically symmetric superposition

$$|\Psi_{TROIKA}^{(2)}\rangle \equiv \frac{1}{\sqrt{2}}(|x\rangle_1 |y\rangle_2 + |y\rangle_1 |x\rangle_2) \equiv \frac{1}{\sqrt{2}}(|xy\rangle + |yx\rangle),$$

which is, up to specific notation, one of the four Bell states—concretely, the symmetric triplet $|\Psi^+\rangle$ of two qubits. The 'enforced trihedron' property reduces here to the statement, already well-known in quantum information, that a measurement of σ_x (or any orthonormal basis) on one of the two particles instantaneously determines the result on the other, in the complementary direction.

6.4. Directional Interaction Hamiltonian: Collapse to Heisenberg XX.

Repeating the construction of Section 2.3 for $d = 2$: we define the discrete phase variable $\theta_i \in \{0, \pi\}$, which instead of a genuine complex phase reduces to a real Ising-type variable,

$$s_i \equiv \cos \theta_i \in \{+1, -1\},$$

since $e^{i\theta_i} = \cos \theta_i$ for $\theta_i \in \{0, \pi\}$ (there is no imaginary part). The cyclically symmetric interaction Lagrangian, analogous to $\mathcal{L}_{int}^{dir} = g \cos(\theta_1 + \theta_2 + \theta_3)$, is written here directly as

$$\mathcal{L}_{int}^{dir,(2)} = g s_1 s_2 = g \cos \theta_1 \cos \theta_2,$$

and, applying the quantization correspondence $s_i = \cos \theta_i \rightarrow \hat{X}_i = \sigma_x^{(i)}$ (the real and more elementary version of the $e^{i\theta_i} \rightarrow \hat{X}_i$ correspondence postulated in section 2.3), we obtain

$$\boxed{\hat{H}_{int}^{dir,(2)} = \hbar g \sigma_x^{(1)} \sigma_x^{(2)}}$$

This expression is, literally, the XX exchange term of an anisotropic Heisenberg model between two spin- $\frac{1}{2}$ particles, and it also coincides with the generator (up to a factor) of the $\sqrt{SWAP}$ logic gate in quantum computing.

Exact Spectrum. Since $\hat{H}_{int}^{dir,(2)}$ is a real and symmetric 4×4 operator, its diagonalization is immediate without the need for complex numbers. Decomposing the basis $\{|xx\rangle, |xy\rangle, |yx\rangle, |yy\rangle\}$ into symmetric/antisymmetric subspaces under source permutation:

$$\hat{H}_{int}^{dir,(2)} \frac{1}{\sqrt{2}}(|xx\rangle \pm |yy\rangle) = \pm \hbar g \frac{1}{\sqrt{2}}(|xx\rangle \pm |yy\rangle),$$

$$\hat{H}_{int}^{dir,(2)} \frac{1}{\sqrt{2}}(|xy\rangle \pm |yx\rangle) = \pm \hbar g \frac{1}{\sqrt{2}}(|xy\rangle \pm |yx\rangle),$$

so that $|\Psi_{TROIKA}^{(2)}\rangle = \frac{1}{\sqrt{2}}(|xy\rangle + |yx\rangle)$ is, once again, the maximum energy eigenstate, $+\hbar g$, doubly degenerate along with $\frac{1}{\sqrt{2}}(|xx\rangle + |yy\rangle)$. The full spectrum is $\{+\hbar g, +\hbar g, -\hbar g, -\hbar g\}$, which exactly preserves the key structural property of the $d = 3$ case, that is: $|\Psi_{TROIKA}\rangle$ occupies the extremal eigenvalue.

6.5. Oscillator-Direction Hamiltonian: Full Derivation with Legendre Transform.

Here, the $d = 2$ case offers a genuine methodological advantage over the general case: since s_i is a real variable (not a complex phase), it is possible to construct the oscillator Lagrangian using the standard real canonical variables, (x_i, p_i) , and carry out a complete and unambiguous Legendre transform. The Lagrangian for source i is:

$$L^{(i)} = \underbrace{\frac{1}{2} m \dot{q}_i^2 - \frac{1}{2} m \omega_i^2 q_i^2}_{\text{free oscillator}} - \underbrace{\lambda' q_i s_i}_{\text{linear, local coupling}}.$$

The interaction term is linear in q_i (the simplest choice compatible with locality and Hermiticity) and does not depend on $\dot{q}_i$, so that the conjugate momentum receives no correction:

$$p_i = \frac{\partial L^{(i)}}{\partial \dot{q}_i} = \dot{q}_i.$$

Applying then the Legendre transform to the previous Lagrangian, we obtain the Hamiltonian function:

$$H^{(i)} = p_i \dot{q}_i - L^{(i)} = \frac{p_i^2}{2m} + \frac{1}{2} m \omega_i^2 q_i^2 + \lambda' q_i s_i.$$

Regarding quantization, we will perform the canonical one (the most standard choice possible: $\hat{q}_i, \hat{p}_i$, with $[\hat{q}_i, \hat{p}_i] = i\hbar$, and $s_i \rightarrow \sigma_x^{(i)}$):

$$\hat{H}^{(i)} = \hbar \omega_i \left(\hat{N}_i + \frac{1}{2} \right) + \lambda' \sqrt{\frac{\hbar}{2m\omega_i}} \sigma_x^{(i)} (\hat{a}_i + \hat{a}_i^\dagger).$$

Absorbing constants into $\lambda \equiv \frac{\lambda'}{\hbar} \sqrt{\frac{\hbar}{2m\omega_i}}$:

$$\boxed{\hat{H}^{(i)} = \hbar \omega_i \left(\hat{N}_i + \frac{1}{2} \right) + \hbar \lambda \sigma_x^{(i)} (\hat{a}_i + \hat{a}_i^\dagger)}$$

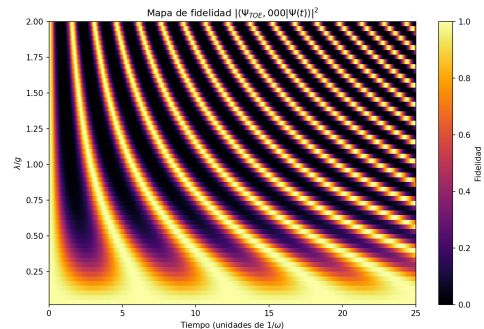
This is, exactly, the **quantum Rabi Hamiltonian**, obtained here not by analogy but through a chain of standard correspondences: ordinary canonical quantization of the oscillator, plus the single postulated correspondence — $s_i \rightarrow \sigma_x^{(i)}$ —, which is the most elementary version possible of the cyclic quantization correspondence used in the general case, as it deals with a group of only two elements.

Following those rules, the complete hamiltonian function for TROIKA system for two sources is:

$$\hat{H}^{(2)} = \sum_{i=1}^2 \hbar \omega_i \left(\hat{N}_i + \frac{1}{2} \right) + \hbar g \sigma_x^{(1)} \sigma_x^{(2)} + \hbar \lambda \sum_{i=1}^2 \sigma_x^{(i)} (\hat{a}_i + \hat{a}_i^\dagger),$$

over the Hilbert space $\mathcal{H}_{TROIKA}^{(2)} = \bigotimes_{i=1}^2 (\mathbb{C}^2 \otimes \mathcal{H}_i^{osc})$, with dimension $[2(N+1)]^2 = 4(N+1)^2$ after truncating —compared to the $27(N+1)^3$ dimensions of the three-source case—, which is another manifestation of the structural simplification of the reduced case.

Attached is a representation of a fidelity map compiled for the three-source system case:



7. Potential Applications and Experimental Viability of the Model.

The formal development of the model, far from being exhausted in its internal mathematical structure, finds a natural echo in several vanguard experimental platforms for quantum computing and quantum information. The fundamental reason is twofold: on the one hand, qutrits (three-level systems) are resources already available and controllable in multiple laboratories; on the other hand, the coupling between discrete (direction) and continuous (oscillation) degrees of freedom proposed by the Hamiltonian $\hat{H} = \hat{H}_0 + \hat{H}_{int}^{dir} + \hat{H}_{int}^{osc-dir}$ reproduces, in essence, the physics of hybrid systems already studied, but with a novel ingredient: the enforced trihedron correlation and the conserved K symmetry. Below, we explore the most promising paths, as well as the challenges they pose.

7.1. Superconducting Circuits.

Superconducting circuits, and in particular **transmons**, have established themselves as the leading platform for superconducting quantum computing. A key, and often underutilized, property is that the transmon is not a native qubit, but rather an **anharmonic Duffing oscillator** with multiple accessible energy levels:

1. **Qutrits without additional hardware:** the first three levels of the transmon can be treated as a qutrit, defining the states $\{|x\rangle, |y\rangle, |z\rangle\}$ as the three internal directions. The community has recently demonstrated full unitary control of qutrits in these devices, as well as the implementation of high-fidelity logic gates specific to qutrits.
2. **Realization of the cyclic coupling:** the term $\hat{H}_{int}^{dir} = \frac{\hbar g}{2} (\hat{X}_1 \hat{X}_2 \hat{X}_3 + h.c.)$ represents a three-body interaction. Although natural couplings in circuits are of the two-body XX or ZZ type, there are consolidated theoretical proposals to induce three-body interactions through parametric pumping or resonator-mediated couplings, modulating the qutrit frequencies to the sum frequencies $\omega_1 + \omega_2 + \omega_3$. Although technically demanding, it is not a fundamental barrier.
3. **Oscillator-qutrit coupling:** the term $\hat{H}_{int}^{osc-dir} = \hbar \lambda \sum_i (\hat{X}_i \hat{a}_i^\dagger + h.c.)$ is directly realizable in the generalized Jaynes-Cummings regime, where a resonator (the oscillator) is capacitively coupled to the cyclic transition of the qutrit. This has already been done with qubits, and extending it to the qutrit is a natural evolution.

In this context, the TROIKA model would serve as a testbed to validate the quality of three-qutrit entanglement. The prediction of the Rabi oscillation $\Omega_{Rabi} =$

$\sqrt{g^2 + (2\lambda)^2}$ and the exact anti-correlation between fidelity and $\langle N_{tot} \rangle$ offer highly precise fitting curves to characterize the noise and decoherence of the system.

7.2. Cold Atoms and Trapped Ions: Control and Scalability.

Ultra-cold neutral atom systems and trapped ions offer exceptional control over both internal (hyperfine) and external (vibrational or motional) degrees of freedom.

1. **Internal states as qutrits:** In alkali atoms such as ^{87}Rb , the hyperfine states of the ground level offer a natural three-dimensional basis with extremely long coherence times. Demonstrations of full qutrit control and entanglement generation in optical lattices have already been achieved.
2. **Coupling to vibrational modes:** In the case of trapped ions, the collective motion modes (stretch) act as bosonic harmonic oscillators, identical to the $\hat{a}_i$ in our model. The $\hat{X}_i \hat{a}_i^\dagger$ coupling is achieved through Raman lasers that excite sideband transitions. This makes trapped ions the most faithful platform to the formalism of this document, as the coupling between spin (or qutrit) and phonons is their standard *modus operandi*.
3. **Simulation of many-body interactions:** Although the present model is restricted to three sources, an optical lattice could host multiple atom trios. Implementing the TROIKA Hamiltonian in each trio would allow studying the competition between fidelity freezing and the exchange regime in a parallelized manner.

7.3. High-Dimensional Quantum Information and Communication.

The state $|\Psi_{TROIKA}\rangle$ is not a mere mathematical artifact; it is an entangled resource with an effective Schmidt rank of 3.

1. **Controlled teleportation:** The cyclic structure $|xyz\rangle + |yzx\rangle + |zxy\rangle$ is identical to that of the states used in dimension-3 quantum state teleportation protocols, where one of the particles acts as a controller. This allows the TROIKA model to serve as a quantum channel to transmit a qutrit from one point to another under the supervision of a third party—a capability impossible to replicate with two qubits (which only offer a Schmidt rank of 2).
2. **Quantum Key Distribution (QKD):** In cryptography, d-dimensional systems are intrinsically more robust against certain attacks due to the larger size of the Hilbert space. The enforced trihedron correlations of the model could be exploited to generate quantum keys with higher tolerated error rates than standard BB84 protocols.

3. **Quantum games and non-locality:** The TROIKA state suggests a natural pathway to explore violations of 3-dimensional Bell inequalities in non-locality games, although this verification lies outside the scope of the present work and will be addressed in future extensions.

7.4. Generation of Non-Classical Oscillator States.

One of the most robust numerical findings of this document is the appearance of negativity in the Wigner function $W(q_1, p_1)$ at the instants of minimal fidelity.

1. **Quantum metrology:** The generation of squeezed or 'Schrödinger cat-type' states in the vibrational modes, induced by the entanglement with the qutrit, is a valuable resource for quantum interferometry. The TROIKA model offers a precise recipe (by adjusting λ/g) to generate these states with a controlled periodicity, without the need for complex non-linear Hamiltonians.
2. **Coherence test:** For the experimentalist, observing the recovery of the initial Gaussian at the fidelity maxima and its deformation at the minima is visual and undeniable proof that the system is not undergoing irreversible decoherence, but rather a pure unitary evolution. This makes the TROIKA model a coherence diagnostic tool for hybrid platforms.

7.5. The Two-Source Limit: A Bridge to Established Physics.

Returning to Section 6, the reduction of the TROIKA model to two sources is not a mere curiosity, but an experimental calibration tool.

By collapsing exactly into the Heisenberg XX Hamiltonian (for the directions) and the quantum Rabi Hamiltonian (for each source), the TROIKA model offers a safety mode for experimentalists. Before attempting the complex three-body coupling, researchers can configure the system in the two-source regime, calibrate the parameters g and λ using the Ω_{Rabi} formula and standard spectroscopy, and then 'turn on' the third source, trusting that the cyclic symmetry will guarantee the conservation of K . This verification hierarchy (2 sources $\rightarrow$ 3 sources) is unusually clear compared to other many-body models.

8. Continuous Extension of the TROIKA Model.

8.1. Geometric Motivation of the Continuous Limit.

Up to this point, the TROIKA model has described the directional degree of freedom of each source through a three-dimensional Hilbert space:

$$\mathcal{H}_i^{dir} = \text{span}\{|x\rangle_i, |y\rangle_i, |z\rangle_i\}.$$

where the three basis states represent three spatially orthogonal orientations.

We know how the cyclic operator defined for each source, $\hat{X}_i$, acts on the basis, and that it satisfies $\hat{X}_i^3 = \hat{\mathbb{I}}_i$.

This structure constitutes a three-dimensional representation of the cyclic group Z_3 and is precisely the internal symmetry that allows us to construct the state:

$$|\Psi_{TROIKA}\rangle = \frac{1}{\sqrt{3}}(|xyz\rangle + |yzx\rangle + |zxy\rangle),$$

which constitutes the fundamental quantum configuration of the discrete model.

However, there is a fundamental geometric question: nothing in the three-dimensional physical space absolutely distinguishes the Cartesian directions x, y, z . They are simply a particular choice of basis. If the model aims to describe a more general geometric structure, it is natural to allow the direction associated with each source to adopt any spatial orientation.

The continuous extension then consists in replacing the discrete label x, y, z with a unit vector $\mathbf{n} \in S^2$, where $S^2 = \{\mathbf{n} \in \mathbb{R}^3 : \mathbf{n} \cdot \mathbf{n} = 1\}$ is the unit sphere.

An orientation can be parameterized by means of two angular coordinates,

$$\theta \in [0, \pi], \varphi \in [0, 2\pi) \text{ such that : } \mathbf{n}(\theta, \varphi) = \begin{pmatrix} \sin \theta \cos \varphi \\ \sin \theta \sin \varphi \\ \cos \theta \end{pmatrix}.$$

We thus transition from a discrete degree of freedom d_i to a continuous variable. This substitution should not be interpreted simply as an alternative parameterization of the same space. The kinematic space changes qualitatively. In the discrete model, there are three possible configurations per source; in the continuous model, there is an infinite family of orientations.

The Hilbert space corresponding to a source therefore changes from being $\mathbb{C}^3$ to a functional space:

$$\mathcal{H}_i^{cont} = L^2(S^2, d\Omega) \text{ with } d\Omega = \sin \theta d\theta d\varphi.$$

A continuous directional state is described by a wave function $\psi_i(\mathbf{n}) = \langle \mathbf{n} | \psi_i \rangle$, and the normalization takes the form $\int_{S^2} d\Omega |\psi_i(\mathbf{n})|^2 = 1$. The resolution of the identity is $\int_{S^2} d\Omega |\mathbf{n}\rangle \langle \mathbf{n}| = \hat{\mathbb{I}}$.

The discrete description can then be recovered as a restriction of the continuous space to three localized directional states. In particular, we can introduce three localized wave functions

$$\psi_x(\mathbf{n}), \quad \psi_y(\mathbf{n}), \quad \psi_z(\mathbf{n}),$$

centered respectively in those directions. The corresponding states

$$|k\rangle = \int_{S^2} d\Omega \psi_k(\mathbf{n}) |\mathbf{n}\rangle, \quad \text{where } k = x, y, z$$

form, when the states are sufficiently localized and their overlaps are negligible, $\langle a|b\rangle \approx \delta_{ab}$, with $a, b \in \{x, y, z\}$.

Thus, the qutrit of the discrete model can be interpreted as the subspace

$$\mathcal{H}_i^{(3)} = \text{span}\{|x\rangle_i, |y\rangle_i, |z\rangle_i\} \subset L^2(S^2, d\Omega).$$

This interpretation is important because it modifies the conceptual status of the qutrit: it is no longer necessarily the fundamental degree of freedom and can be understood as an effective low-energy sector of a continuous directional theory.

Consequently, the discrete model should be considered a particular realization of a more general theory where the fundamental variable is the orientation $\mathbf{n}_i \in S^2$. The real problem then consists in determining which simultaneous configurations $(\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3)$ are compatible with the geometric structure that defines the TROIKA model.

8.2. Direction as a Continuous Variable over S^2 .

Let us now consider the three sources F_1, F_2, F_3 . We associate a unit vector with each one: $\mathbf{n}_i \in S^2$, $i = 1, 2, 3$. The directional configuration space without imposing any relationship between the sources is

$$\mathcal{C}_0 = S^2 \times S^2 \times S^2 = (S^2)^3.$$

Each configuration is given by three angular pairs

$$(\theta_1, \varphi_1), (\theta_2, \varphi_2), (\theta_3, \varphi_3).$$

In the absence of constraints, the system therefore possesses six independent continuous coordinates.

The scalar product between two directions is given by $\mathbf{n}_i \cdot \mathbf{n}_j = \cos \gamma_{ij}$, where γ_{ij} is the angle formed by both orientations.

In particular, $\mathbf{n}_i \cdot \mathbf{n}_j = 0$ is equivalent to $\gamma_{ij} = \frac{\pi}{2}$. The fundamental property that characterizes the TROIKA trihedron is precisely that the three sources must be oriented along three mutually orthogonal directions. Therefore, the physically relevant configurations must satisfy

$$\mathbf{n}_i \cdot \mathbf{n}_j = 0, \quad \forall i \neq j$$

Along with the individual normalization, $\mathbf{n}_i \cdot \mathbf{n}_i = 1$, these conditions can be written in a single matrix equation.

Let us then define the matrix

$$R = \begin{pmatrix} | & | & | \\ \mathbf{n}_1 & \mathbf{n}_2 & \mathbf{n}_3 \\ | & | & | \end{pmatrix}.$$

Then the components of the product $R^T R$ are: $(R^T R)_{ij} = \mathbf{n}_i \cdot \mathbf{n}_j$. Therefore, the orthonormality conditions are exactly equivalent to $R^T R = \mathbb{I}_3$. This implies that $R \in O(3)$, the group of three-dimensional orthogonal transformations.

Up to this point, an ambiguity still remains: an orthogonal matrix can represent either a rotation or a transformation that includes an orientation reversal. To distinguish between both cases, we must consider the determinant.

From $R^T R = \mathbb{I}_3$, it follows that $(\det R)^2 = 1$, which implies that necessarily $\det R = \pm 1$.

To describe a trihedron oriented in the same way as the reference Cartesian trihedron, we require $\det R = +1$. Equivalently,

$$\boxed{\det R = +1 \implies \mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{n}_3}$$

This condition selects right-handed trihedrons and eliminates reference frames with inverted orientation.

The allowed configuration set is then given by

$$\mathcal{C}_{TROIKA} = \{(\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3) \in (S^2)^3 : \mathbf{n}_i \cdot \mathbf{n}_j = \delta_{ij}, \mathbf{n}_1 \times \mathbf{n}_2 = \mathbf{n}_3\}$$

This has an especially simple geometric interpretation. Once the first direction, $\mathbf{n}_1 \in S^2$, is chosen arbitrarily, the second direction must belong to the plane perpendicular to $\mathbf{n}_1$. The intersection of that plane with the unit sphere is a circle,

$$\{\mathbf{n}_2 \in S^2 : \mathbf{n}_1 \cdot \mathbf{n}_2 = 0\} \simeq S^1.$$

Therefore, once $\mathbf{n}_1$ is determined, there is still a single angular degree of freedom associated with the rotation of $\mathbf{n}_2$ around $\mathbf{n}_1$.

Finally, the third direction is completely determined by $\mathbf{n}_3 = \mathbf{n}_1 \times \mathbf{n}_2$. Consequently, the total number of degrees of freedom is $2 + 1 = 3$. This count matches exactly the dimension of $SO(3)$, $\dim SO(3) = 3$.

This suggests that the configuration space of oriented orthogonal trihedrons is not merely an arbitrary subset of $(S^2)^3$, but rather possesses a group structure.

8.3. The Configuration Space of the Trihedron: $SO(3)$.

Let us take the Cartesian trihedron $\mathcal{E} = (\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$ as a reference frame. For any admissible configuration $(\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3)$, there exists a linear transformation R such that

$$\mathbf{n}_1 = R\mathbf{e}_x, \quad \mathbf{n}_2 = R\mathbf{e}_y, \quad \mathbf{n}_3 = R\mathbf{e}_z.$$

In matrix form, the conditions

$$R^T R = \mathbb{I}_3, \quad \det R = +1 \implies R \in SO(3).$$

Therefore, there is a one-to-one correspondence between the configurations of an oriented trihedron and the elements of $SO(3)$:

$$\boxed{\mathcal{C}_{TROIKA} \simeq SO(3)}$$

In the discrete model, the three sources possess the orientations

$$(\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z), (\mathbf{e}_y, \mathbf{e}_z, \mathbf{e}_x), (\mathbf{e}_z, \mathbf{e}_x, \mathbf{e}_y).$$

The first of these configurations corresponds to $R_0 = \mathbb{I}_3$. The other two are obtained by a rotation of $2\pi/3$ around the axis

$$\hat{\mathbf{u}} = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

Let $R_3 = R_{\hat{\mathbf{u}}}(\frac{2\pi}{3})$; by construction,

$$R_3 \mathbf{e}_x = \mathbf{e}_y, \quad R_3 \mathbf{e}_y = \mathbf{e}_z, \quad R_3 \mathbf{e}_z = \mathbf{e}_x$$

Furthermore, $R_3^3 = \mathbb{I}_3$. The set $\{\mathbb{I}_3, R_3, R_3^2\}$ forms, therefore, the cyclic subgroup

$$\boxed{C_3 \simeq Z_3 \subset SO(3)}$$

The operator X_i should not be interpreted then as a geometric transformation completely independent of physical space. Its action on the discrete states represents, in the continuous limit, the rotation action of R_3 on the reference trihedron.

Consequently,

$$|x\rangle \rightarrow |y\rangle \rightarrow |z\rangle \rightarrow |x\rangle$$

constitutes the discretized version of

$$\mathbf{e}_x \rightarrow \mathbf{e}_y \rightarrow \mathbf{e}_z \rightarrow \mathbf{e}_x$$

induced by an element of $SO(3)$ of order three.

The fundamental difference is that in the discrete model only the three elements of the subgroup C_3 are considered, whereas the continuous theory allows any $R \in SO(3)$. The transition can be visualized schematically as $Z_3 \subset SO(3)$, where the cyclic structure constitutes only a discrete sector within the full space of orientations.

Interpretation via Euler Angles:

Another equivalent way to express this structure consists in parameterizing R through three real parameters. For instance, using Euler Angles,

$$R = R_x(\alpha)R_y(\beta)R_z(\gamma), \quad \text{with } \alpha, \gamma \in [0, 2\pi), \quad \beta \in [0, \pi].$$

Thus, the collective orientation of the complete system can be expressed via a single coordinate $R(t) \in SO(3)$, instead of three independent directional variables.

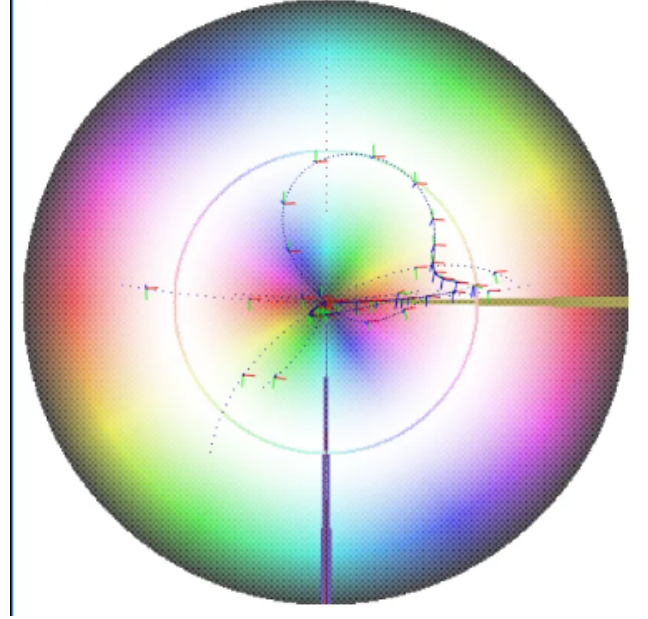


Figure 1: Simple representation of the three-dimensional rotation group $SO(3)$

The three orientations $\mathbf{n}_1(t), \mathbf{n}_2(t), \mathbf{n}_3(t)$ do not need to evolve independently if the orthogonal trihedron is preserved during the dynamics. It is sufficient to introduce a time-dependent rotation matrix, $R(t) \in SO(3)$, and define:

$$\mathbf{n}_1(t) = R(t)\mathbf{e}_x, \quad \mathbf{n}_2(t) = R(t)\mathbf{e}_y, \quad \mathbf{n}_3(t) = R(t)\mathbf{e}_z.$$

The orthogonality is then automatically guaranteed for all t :

$$\mathbf{n}_i(t) \cdot \mathbf{n}_j = \mathbf{e}_i^T R(t)^T R(t) \mathbf{e}_j = \mathbf{e}_i^T \mathbf{e}_j = \delta_{ij}.$$

In this way, the geometric constraint of the model ceases to be a condition that must be continuously imposed on three independent variables, and instead becomes directly incorporated into the parameterization of the configuration space.

This formulation finally allows us to introduce the **angular velocity of the trihedron**. Since $R^T R = \mathbb{I}_3$, differentiating with respect to time yields:

$$\frac{d}{dt}(R^T R) = \dot{R}^T R + R^T \dot{R} = 0,$$

Therefore, the matrix

$$\Omega = R^T \dot{R}$$

is skew-symmetric: $\Omega^T = -\Omega$.

Any 3×3 skew-symmetric matrix can be written as:

$$\Omega = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix},$$

which we can identify with the matrix operator associated with the angular velocity vector:

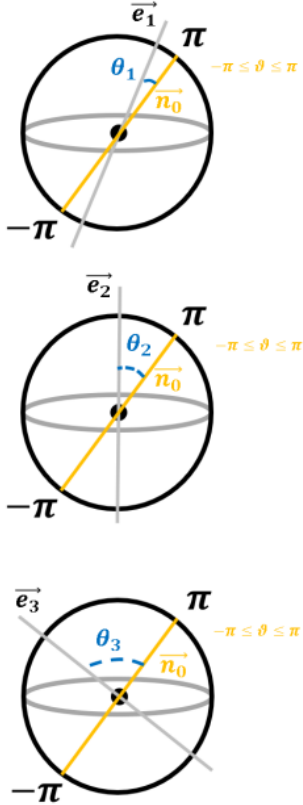
$$\omega = \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}.$$

The evolution of the orientation is thus determined by:

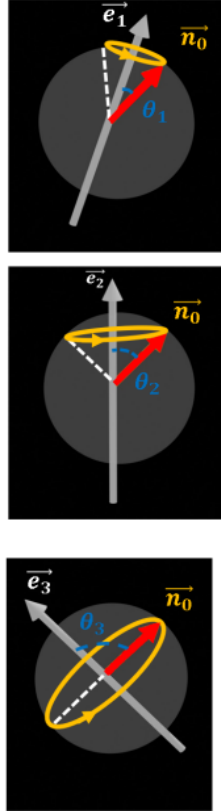
$$R^T \dot{R} = \hat{\omega}$$

This proves that the collective degree of freedom of the continuous TROIKA possesses exactly the kinematic structure of a free rigid body: its configuration is an element of $SO(3)$ and its velocity is an element of the Lie algebra $\mathfrak{so}(3)$.

Lie algebra $\mathfrak{so}(3)$ π -ball



Lie group S_2



The continuous formulation of TROIKA is then summarized by the geometric chain:

$$\text{discrete qutrit} \rightarrow S^2 \rightarrow \mathcal{C}_{TROIKA} \simeq SO(3)$$

while the original cyclic structure satisfies $Z_3 \simeq C_3 \subset SO(3)$.

8.4. Classical Lagrangian of the Directional Sector.

The most general form of rotational kinetic energy corresponds to a rigid rotor—for instance, the case of a top as a rigid body—which is expressed as:

$$T_{rot} = \frac{1}{2} \omega^T \mathbb{I} \omega,$$

where $\mathbb{I}$ is the inertia tensor. In the principal axes of the system, the products of inertia vanish, leaving only the principal moments of inertia:

$$\mathbb{I} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}, \text{ such that } T_{rot} = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2).$$

However, the TROIKA model does not originally introduce any privileged spatial direction. The initial construction treats all three directions symmetrically. Therefore, the most natural choice for a fundamental theory is the isotropic case, $I_1 = I_2 = I_3 = I$. Then,

$$T_{rot} = \frac{I}{2} (\omega_1^2 + \omega_2^2 + \omega_3^2) = \frac{I}{2} |\vec{\omega}|^2.$$

The free Lagrangian is therefore: $L_{rot} = \frac{I}{2} |\vec{\omega}|^2$.

8.4.1. Introduction of the TROIKA Interaction.

In the strict formulation we have just constructed, where $R \in SO(3)$, the trihedron is orthogonal by construction.

However, we can also consider a more general theory in which we initially allow three independent vectors,

$$(\mathbf{n}_1, \mathbf{n}_2, \mathbf{n}_3) \in (S^2)^3,$$

and let the orthogonality emerge dynamically.

In that case, we can introduce the potential:

$$V_{ort} = \frac{\kappa}{2} [(\mathbf{n}_1 \cdot \mathbf{n}_2)^2 + (\mathbf{n}_2 \cdot \mathbf{n}_3)^2 + (\mathbf{n}_3 \cdot \mathbf{n}_1)^2], \quad \kappa > 0$$

This potential automatically satisfies $V_{ort} \geq 0$; furthermore:

$$V_{ort} = 0 \iff \mathbf{n}_1 \cdot \mathbf{n}_2 = \mathbf{n}_2 \cdot \mathbf{n}_3 = \mathbf{n}_3 \cdot \mathbf{n}_1 = 0$$

Although I will not employ this construction in the present document, this in-depth analysis is left as a possible extension to study excitations outside the orthogonal subspace.

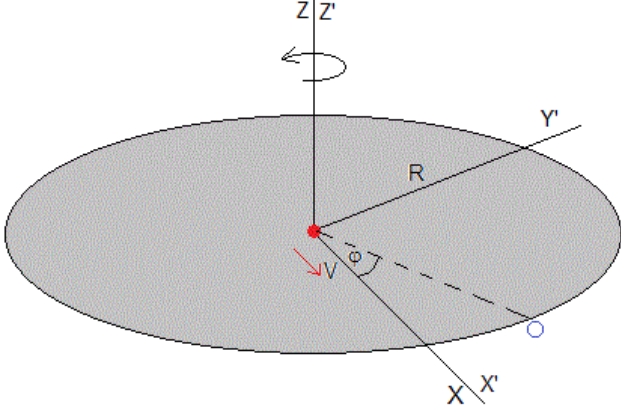
8.4.2. Angular Momentum and Equations of Motion.

Starting from $L_{dir} = \frac{I}{2} |\vec{\omega}|^2$, the body-fixed angular momentum is:

$$\mathbf{L} = \frac{\partial L_{dir}}{\partial \omega} = I \omega.$$

For the free isotropic rotor, there are no external torques, and therefore $\frac{d\mathbf{L}}{dt} = 0$. Thus, $\dot{\omega} = 0$ in the space frame. This means that, classically, the configuration of the trihedron evolves as a uniform rotation.

See the following schematic representation of a uniform rotational motion of the configuration, where $\omega \sim \text{const.}$:



The solution can be written formally as: $R(t) = R(0) \exp(t\hat{\omega})$, when the angular velocity is constant. This demonstrates that the trajectory belongs to $SO(3)$ at all times; that is, the system does not escape the space of orthogonal trihedrons during its evolution.

8.5. Quantization of the Rotational Sector.

In the classical theory, the configuration variable is $R \in SO(3)$. Therefore, the wave function does not depend on ordinary Cartesian coordinates, but rather on the group element: $\Psi = \Psi(R)$. The natural Hilbert space is

$$\mathcal{H}_{rot} = L^2(SO(3), d\mu_H),$$

where $d\mu_H$ is the Haar measure of the group.

Definition (Haar measure): In any compact topological group G , there exists a unique regular Borel probability measure m , which is invariant under left (right) translations, in the sense that:

1. $\int_G f dm = \int_G (L_s f) dm$, with $s \in G$, $f \in C(G)$
2. $\int_G f dm = \int_G (R_s f) dm$, with $s \in G$, $f \in C(G)$
3. $\int_G f(x) dm(x) = \int_G f(x^{-1}) dm(x)$, $f \in C(G)$

This m is the so-called Haar measure.

The normalization of a state is then

$$\int_{SO(3)} d\mu_H(R) |\Psi(R)|^2 = 1.$$

This is the natural generalization of the normalization of a wave function over an ordinary configuration space.

8.5.1. Angular Momentum Operators.

The infinitesimal algebra of $SO(3)$ is $\mathfrak{so}(3)$. Its three quantum generators are represented by the operators $\hat{J}_1, \hat{J}_2, \hat{J}_3$, which satisfy:

$$[\hat{J}_i, \hat{J}_j] = i\hbar \epsilon_{ijk} \hat{J}_k,$$

where ϵ_{ijk} is the Levi-Civita tensor. We define the **Casimir Operator**

$$\hat{\mathbf{J}}^2 = \hat{J}_1^2 + \hat{J}_2^2 + \hat{J}_3^2.$$

As in the classical rotor $H_{rotor} = \frac{\mathbf{L}^2}{2I}$, canonical quantization leads to $\hat{H} = \frac{1}{2I} \hat{\mathbf{J}}^2$. This is, therefore, the fundamental expression of the continuous rotational Hamiltonian.

In the coordinate representation over the group itself, the Casimir operator is related to the Laplace-Beltrami operator:

$$\hat{\mathbf{J}}^2 = -\hbar^2 \Delta_{SO(3)}.$$

Definition (Laplace-Beltrami Operator): The Laplace-Beltrami operator extends the notions of divergence and gradient to curved spaces equipped with a metric (it is a generalization of the Laplace operator to Riemannian or pseudo-Riemannian manifolds).

Formally, let (M, g) be an oriented Riemannian or pseudo-Riemannian manifold of dimension n , with a metric tensor g_{ij} . The Laplace-Beltrami operator Δ acts on smooth functions $f : M \rightarrow \mathbb{R}$ as the divergence of the gradient: $\Delta f = \text{div}(\text{grad}(f))$. In a local coordinate system x^i , it is explicitly defined by:

$$\Delta f = \frac{1}{\sqrt{|g|}} \sum_{i,j=1}^n \frac{\partial}{\partial x^i} \left(g^{ij} \sqrt{|g|} \frac{\partial f}{\partial x^j} \right)$$

Definition (Casimir Operator): It is a key algebraic tool in the representation theory of Lie algebras and Lie groups.

Formally, let $\mathfrak{g}$ be a finite-dimensional Lie algebra over a field of characteristic zero (such as $\mathbb{R}$ or $\mathbb{C}$) and $\mathcal{U}(\mathfrak{g})$ its universal enveloping algebra. The Casimir operator is an element $C \in \mathcal{U}(\mathfrak{g})$ that belongs to the center of $\mathcal{U}(\mathfrak{g})$, which means that it commutes with every element of the Lie algebra: $[C, X] = 0$, for all $X \in \mathfrak{g}$.

Mathematically, if $\mathfrak{g}$ is a semisimple Lie algebra, the Killing form $B(X, Y) = \text{trace}(ad(X)ad(Y))$ is used. Let $\{X_a\}$ be a basis of $\mathfrak{g}$ and $\{X^a\}$ be the dual basis with respect to the Killing form ($B(X_a, X^b) = \delta_a^b$). The Casimir operator is defined as:

$$C = \sum_a X_a X^a$$

With all that said, we obtain the following equation:

$$\hat{H}_{rot} = -\frac{\hbar^2}{2I} \Delta_{SO(3)}.$$

This equation tells us that the directional Hamiltonian of the continuous TROIKA is the Laplacian over the configuration space $SO(3)$. Therefore, we have a quantum particle living on the configuration manifold of all orthogonal trihedrons.

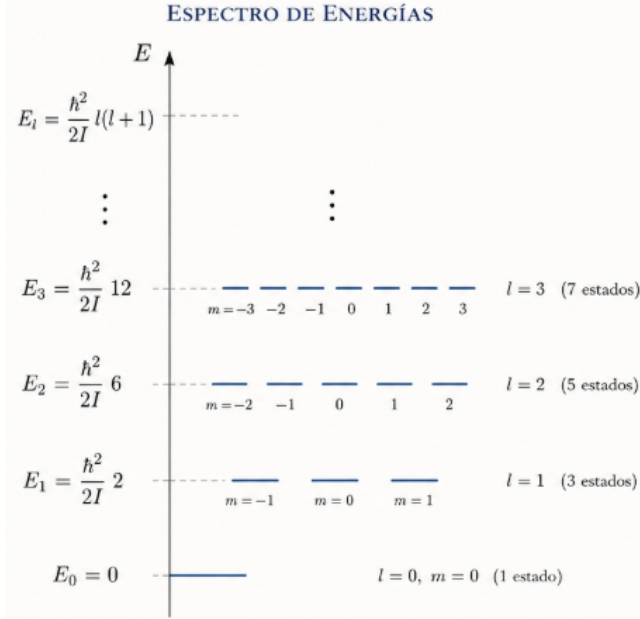
8.5.2. Spectrum of the Rotor.

The eigenfunctions of $\hat{\mathbf{J}}^2$ are associated with the irreducible representations of $SO(3)$. They are labeled by $j = 0, 1, 2, \dots$, and satisfy:

$$\hat{\mathbf{J}}^2 |j, m, \nu\rangle = \hbar^2 j(j+1) |j, m, \nu\rangle.$$

Consequently, $E_j = \frac{\hbar^2}{2I} j(j+1)$. The rotational levels exhibit the characteristic structure:

$$E_0 = 0, E_1 = \frac{\hbar^2}{I}, E_2 = \frac{3\hbar^2}{I}, \dots$$



The full degeneracy of the rotor over $SO(3)$ is $(2j+1)^2$, since the $SO(3)$ space possesses two independent actions of left and right rotation.

8.5.3. Recovery of the Z_3 Structure.

In the discrete description, each source possesses the operator $\hat{X}_i$, which satisfies $\hat{X}_i^3 = \mathbb{I}$ and performs the transformation:

$$|x\rangle \rightarrow |y\rangle \rightarrow |z\rangle \rightarrow |x\rangle.$$

In the continuous regime, this transformation is geometrically identified with the rotation $R_3 = R_{\hat{\mathbf{u}}}(\frac{2\pi}{3})$ around the axis $\hat{\mathbf{u}} = \frac{1}{\sqrt{3}}(1, 1, 1)$. This rotation satisfies $R_3^3 = \mathbb{I}$. Therefore, it generates the subgroup:

$$\langle R_3 \rangle = \{\mathbb{I}, R_3, R_3^2\} \simeq Z_3.$$

Thus, the discrete cyclic symmetry of the model does not vanish in the continuous limit. It becomes a discrete symmetry contained within the continuous rotational symmetry: $Z_3 \subset SO(3)$. This relationship is what we need to interpret the qutrit as a discrete reduction of the orientation space.

$$\boxed{\text{qutrit} \rightarrow S^2 \rightarrow SO(3) \rightarrow L^2(SO(3))}$$

The directional degree of freedom of the TROIKA model can be interpreted as a collective quantum rotor whose configuration space is the space of all oriented orthonormal trihedrons.

8.6. Coupling Between the Rotational Degree of Freedom and the Harmonic Oscillator.

Each source F_i now possesses two physically distinct degrees of freedom:

1. a rotational degree of freedom, described by a configuration $R_i \in SO(3)$.
2. a vibrational degree of freedom, described by a harmonic oscillator.

The classical local configuration space of a source can therefore be written as $SO(3) \times \mathbb{R}$, if we represent the oscillator through a coordinate q_i .

The objective is to construct an interaction that allows the coherent exchange of excitation between both sectors, maintaining the philosophy of the discrete term of the original model:

$$\hat{H}_{osc-dir} = \hbar\lambda \left(\hat{X}_i \hat{a}_i^\dagger + \hat{X}_i^\dagger \hat{a}_i \right).$$

The central question is, therefore, to determine what continuous operator replaces $\hat{X}_i$.

8.6.1. Translation Operator over $SO(3)$.

In the discrete model, $\hat{X}_i$ does not internally modify the amplitude of a state, but rather shifts its directional label within a cycle. In the continuous regime, the natural generalization of this operation is a finite translation over the rotational configuration space.

Let $G \in SO(3)$ be a fixed rotation. We define the operator $\hat{U}_G$ through its action on a rotational wave function:

$$(\hat{U}_G \Psi)(R) = \Psi(G^{-1}R).$$

Equivalently, on the configuration states:

$$\hat{U}_G |R\rangle = |GR\rangle.$$

The transformation $R \rightarrow GR$ represents an additional rotation of the trihedron. Since $G \in SO(3)$, the operation preserves the Haar measure and, therefore, $\hat{U}_G$ is unitary:

$$\hat{U}_G^\dagger \hat{U}_G = \hat{U}_G \hat{U}_G^\dagger = \mathbb{I}.$$

and furthermore, it holds in particular that: $\hat{U}_G^\dagger = \hat{U}_{G^{-1}}$.

8.6.2. Recovery of the Discrete Cyclic Operator.

We can now verify that the previous operator contains the operator $\hat{X}_i$ as a particular case. Let us take the rotation: $G_3 = R_{\hat{\mathbf{u}}}(\frac{2\pi}{3})$, with $\hat{\mathbf{u}} = \frac{1}{\sqrt{3}}(1, 1, 1)$.

We know that:

$$G_3 \mathbf{e}_x = \mathbf{e}_y, \quad G_3 \mathbf{e}_y = \mathbf{e}_z, \quad G_3 \mathbf{e}_z = \mathbf{e}_x$$

Therefore, in the discrete subspace generated by the three localized states around the Cartesian axes:

$$\hat{U}_{G_3} |x\rangle = |y\rangle, \quad \hat{U}_{G_3} |y\rangle = |z\rangle, \quad \hat{U}_{G_3} |z\rangle = |x\rangle.$$

The restriction of $\hat{U}_{G_3}$ to the qutrit subspace is, therefore:

$$\hat{U}_{G_3}|_{\mathcal{H}(3)} = \hat{X} \implies \hat{X} = P_3 \hat{U}_{G_3} P_3$$

where P_3 is the projector onto the three-dimensional subspace of the discrete model.

$$\text{continuous translation over } SO(3) \rightarrow \text{qutrit operator } X$$

Consequently, the operator X_i is no longer fundamental, as it appears as an effective representation of a continuous transformation.

8.6.3. Infinitesimal Form and Angular Momentum.

An additional advantage of this construction is that we can study the limit of arbitrarily small rotational transformations. Let us take a rotation around an axis $\hat{\mathbf{n}}$ by a small angle $\delta\phi$. Its unitary representation can be written as:

$$\hat{U}(\delta\phi) = \exp\left(-\frac{i}{\hbar}\delta\phi\hat{\mathbf{n}} \cdot \hat{\mathbf{J}}\right),$$

where $\hat{\mathbf{J}}$ is the angular momentum operator.

For $\delta\phi \ll 1$:

$$\hat{U}(\delta\phi) \simeq \mathbb{I} - \frac{i}{\hbar}\delta\phi\hat{\mathbf{n}} \cdot \hat{\mathbf{J}}.$$

Therefore, the angular momentum operator constitutes the infinitesimal generator of the transformations that, at a finite scale, we perform through $\hat{U}_G$; this establishes a direct relationship between:

$$SO(3) \iff \mathfrak{so}(3) \iff \hat{\mathbf{J}}.$$

8.7. Quantization of the Harmonic Oscillator.

Let us move on to the second degree of freedom of each source. The quantization method for the harmonic oscillator will be exactly identical to the one presented in Section 3, meaning that we establish the correspondence oscillator amplitude $\iff$ creation / annihilation operators, ultimately obtaining:

$$\hat{H}_{osc} = \sum_{i=1}^3 \hbar\omega_i \left(\hat{a}_i^\dagger \hat{a}_i + \frac{1}{2} \right).$$

8.7.1. Construction of the Coupling.

We now seek an interaction that allows the coherent exchange of excitation between the rotational and vibrational sectors. The most direct structure, following the original discrete Hamiltonian, is:

$$\hat{H}_{rot-osc} = \hbar \sum_{i=1}^3 \lambda_i \left(\hat{U}_i \hat{a}_i^\dagger + \hat{U}_i^\dagger \hat{a}_i \right)$$

This expression is Hermitian, because:

$$(\hat{U}_i \hat{a}_i^\dagger)^\dagger = \hat{a}_i \hat{U}_i^\dagger = \hat{U}_i^\dagger \hat{a}_i,$$

since the rotational and vibrational operators act on different Hilbert spaces and, therefore:

$$[\hat{U}_i, \hat{a}_j] = [\hat{U}_i, \hat{a}_j^\dagger] = 0 \implies \hat{H}_{rot-osc}^\dagger = \hat{H}_{rot-osc}.$$

Physical Interpretation:

The term $\hat{U}_i \hat{a}_i^\dagger$ creates a vibrational quantum and, simultaneously, applies a rotational transformation to the orientation state. On the other hand, $\hat{U}_i^\dagger \hat{a}_i$ destroys a vibrational quantum and applies the inverse rotational transformation.

In particular, in the continuous version, if we choose $\hat{U}_i = \hat{U}_{G_3}$, we immediately recover:

$$\hat{H}_{rot-osc} \rightarrow \hbar \sum_i \lambda_i \left(\hat{X}_i \hat{a}_i^\dagger + \hat{X}_i^\dagger \hat{a}_i \right)$$

8.7.2. Formulation in Terms of Rotational Generators.

If we wish to work in the small-rotations regime, we write:

$$\hat{U}_i = \exp\left(-\frac{i}{\hbar}\delta\phi\hat{\mathbf{n}} \cdot \hat{\mathbf{J}}\right) \xrightarrow[\delta\phi_i \ll 1]{} \mathbb{I} - \frac{i}{\hbar}\delta\phi\hat{\mathbf{n}} \cdot \hat{\mathbf{J}}.$$

then, the adjoint of this operator will be:

$$\hat{U}_i^\dagger = \mathbb{I} + \frac{i}{\hbar}\delta\phi\hat{\mathbf{n}} \cdot \hat{\mathbf{J}}.$$

Substituting into the rotation-oscillator interaction Hamiltonian:

$$\hat{H}_{rot-osc} \simeq \hbar \sum_i \lambda_i \left[\hat{a}_i^\dagger + \hat{a}_i + \frac{i\delta\phi_i}{\hbar} (\hat{a}_i - \hat{a}_i^\dagger) \hat{\mathbf{n}}_i \cdot \hat{\mathbf{J}}_i \right]$$

Since the first term, $\hbar\lambda_i(\hat{a}_i + \hat{a}_i^\dagger)$, is a static displacement of the oscillator, it can be eliminated by redefining the vibrational equilibrium. Therefore, to first order, the dynamically relevant term becomes:

$$\hat{H}_{rot-osc} \sim i \sum_i \lambda_i \delta\phi_i (\hat{a}_i - \hat{a}_i^\dagger) \hat{\mathbf{n}}_i \cdot \hat{\mathbf{J}}_i$$

hence, in the continuous regime of small rotations, the coupling between vibration and direction turns into a coupling between the conjugate vibrational quadrature and the rotational angular momentum.

8.8. Total Hilbert Space in the Continuum and Full Quantized Hamiltonian.

We can now construct the Hilbert space of the complete continuous model. For each source, we have:

$$\mathcal{H}_i = \mathcal{H}_{rot,i} \otimes \mathcal{H}_{osc,i}$$

where, in particular, $\mathcal{H}_{rot,i} = L^2(SO(3))$ and $\mathcal{H}_{osc,i} = \text{span}\{|0\rangle_i, |1\rangle_i, |2\rangle_i, \dots\}$. For the three sources, it becomes:

$$\mathcal{H}_{TROIKA}^{cont} = \bigotimes_{i=1}^3 [L^2(SO(3)) \otimes \mathcal{H}_{osc,i}]$$

With all that said, we can finally piece together the total Hamiltonian. The vibrational sector is:

$$\hat{H}_{osc} = \sum_{i=1}^3 \hbar \omega_i \left(\hat{a}_i^\dagger \hat{a}_i + \frac{1}{2} \right).$$

The free rotational sector is given by:

$$\hat{H}_{rot} = \sum_{i=1}^3 \frac{1}{2I_i} \hat{\mathbf{J}}_i^2,$$

for the case of independent isotropic rotors. The local direction-vibration interaction is:

$$\hat{H}_{rot-osc} = \hbar \sum_{i=1}^3 \lambda_i \left(\hat{U}_i \hat{a}_i^\dagger + \hat{U}_i^\dagger \hat{a}_i \right).$$

Lastly, we have the interaction between the three orientations. The most straightforward construction in the continuum is an interaction that favors orthogonal configurations and, when reduced to the discrete sector, reproduces the cyclic structure of TROIKA. We can represent it generally by means of an operator $\hat{V}_{TROIKA}(R_1, R_2, R_3)$.

Therefore, the general continuous Hamiltonian can be written as:

$$\begin{aligned} \hat{H}_{TROIKA}^{cont} = & \underbrace{\sum_{i=1}^3 \frac{\hat{\mathbf{J}}_i^2}{2I_i}}_{\text{cont. rotational dyn.}} + \underbrace{\sum_{i=1}^3 \hbar \omega_i \left(\hat{N}_i + \frac{1}{2} \right)}_{\text{quantum vibration}} + \\ & + \underbrace{\hat{V}_{TROIKA}}_{\text{geometric correlation between sources}} + \underbrace{\hbar \sum_{i=1}^3 \lambda_i \left(\hat{U}_i \hat{a}_i^\dagger + \hat{U}_i^\dagger \hat{a}_i \right)}_{\text{vibration-rotation exchange}} \end{aligned}$$

With this, I conclude the analysis and development of the present model. For future works, what remains is to explicitly construct the potential $\hat{V}_{TROIKA}$ so that it is not merely abstract: we want it to have a continuous geometric form, to be $SO(3)$ -invariant, and, when projected onto the Z_3 subspace, to yield exactly:

$$\frac{\hbar g}{2} \left(\hat{X}_1 \hat{X}_2 \hat{X}_3 + \hat{X}_1^\dagger \hat{X}_2^\dagger \hat{X}_3^\dagger \right)$$

9. Conclusions.

In this work, the TROIKA (Triple Rotor Oscillator Interlinked with K-cycle Architecture) model has been developed, presenting a quantum construction based on three coupled sources, each characterized by a three-dimensional directional degree of freedom and a harmonic oscillator. A cyclically entangled state representing an orthogonal trihedron has been defined, and the collective operator $X_1 X_2 X_3$ has been constructed, whose Z_3 symmetry determines the fundamental structure of the directional sector.

The incorporation of the oscillators allows for the study of the coherent exchange between the directional and vibrational degrees of freedom. The numerical analysis reveals two main regimes: a protection regime, for $\lambda \ll g$, where the TROIKA state remains practically stable, and a coherent Rabi-like exchange regime for comparable couplings. Furthermore, the decomposition into symmetry sectors shows that the dynamics remain confined to the sector compatible with the symmetry of the initial state.

The analysis of the reduced density matrix, the entanglement entropy, and the Wigner functions confirms that the temporary loss of fidelity does not correspond to an irreversible decoherence, but rather to a unitary and reversible transfer of coherence toward the oscillators, accompanied by the generation of non-classical vibrational states.

As an additional validation, the two-source limit reproduces well-known structures, including the Heisenberg-type exchange interaction and the quantum Rabi Hamiltonian. Finally, a continuous extension based on rotational degrees of freedom has been formulated, where a geometric potential selects orthogonal configurations related by C_3 , and the tunneling between them allows for the interpretation of the $X_1 X_2 X_3$ term as an effective low-energy interaction. The model thus provides a unified framework for studying geometric entanglement, vibrational dynamics, and discrete symmetries, leaving open future extensions toward more general rotational and continuous formulations.